

Coverage versus Selectivity in Pre-Export Row Scaling for Generated Mixed-Integer Programs

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Abstract

Generated mixed-integer programs (MIPs) reach the solver as flat coefficient matrices, although the generator that builds them knows which rows belong to each component and which rows couple components. We ask whether this pre-export metadata can usefully drive positive row scaling—a transformation that multiplies complete constraints by strictly positive factors and therefore preserves the mathematical program exactly, while numerical feasibility of the returned solutions still has to be verified in the original scale. We propose a structure-aware row-scaling layer that applies a per-row geometric-mean normalization ordered by role (local rows, then block scales, then coupling rows) and evaluate it on a hydrothermal-dispatch generator against Ruiz-kernel baselines within the same pipeline, the solvers' internal scaling, and a fully role-blind control that applies the same kernel to every row, across three instance classes, two commercial solvers, and two tolerances. The metadata acts as a selection policy that decides *where* to normalize, and its value proves class- and solver-dependent: structure-aware variants are associated with substantially lower path-specific solver-facing conditioning diagnostics than the unscaled model and the Ruiz-kernel baselines on the operational testbed—though no general advantage over the role-blind kernel is found—and the runtime gains under Gurobi flow from the per-row kernel itself, with full coverage beating role selectivity on some classes while the reported most-integrated campaign shows a directionally consistent reversal, and no robust runtime gain appears under CPLEX. Under intent-to-treat timing the Gurobi gain appears in all three classes; after penalizing runs whose returned solutions fail original-scale verification, the robust runtime advantage remains clearest in SG-Ter-Mer and does not survive in Simple or Full. The main limitation is a single generated platform: the coverage-versus-selectivity trade-off must be measured per class and per solver rather than assumed from structure.

KEYWORDS

Numerical scaling, mixed-integer programming, generated formulations, block-structured optimization, engineering optimization, hydrothermal dispatch

PRACTITIONER POINTS

- Model generators know each row's role, block ownership, and coupling incidence—metadata a flat export discards but that can drive numerical preprocessing.
- Positive, structure-keyed row factors applied in a local–block–coupling order improved the solver-facing representation on the operational testbed without changing the MIP; on an already well-scaled model the selective rule can widen the exported range instead.
- The runtime lever is per-row normalization; whether roles should gate it is class-dependent. A role-blind control ties the structure-aware rule on the simplest class, beats it on the coupled class (the role-based rule missed the worst-scaled rows), and is directionally slower in the reported most-integrated campaign—so coverage and selectivity must be measured, not assumed.

- The runtime benefit is solver- and reliability-dependent: under intent-to-treat timing, gains appear in the Gurobi campaigns; after penalizing runs whose returned solutions fail original-scale verification, the robust advantage remains clearest in SG-Ter-Mer; and no robust gain appears under CPLEX. Scaling gains should be validated per solver and checked in the original scale, not assumed.

1 | INTRODUCTION

A model generator and a solver see different representations of the same optimization problem. The generator builds the model from components: hydro plants, thermal units, market rules, resource limits, demand balances, and discrete operating decisions. While doing so, it knows which rows are local to one component, which variables that component owns, and which constraints link several components. The solver receives the exported version: rows, columns, coefficients, bounds, and tolerances. In that exported matrix, much of the generator’s structural knowledge has disappeared.

This loss matters because numerical representation affects computation. Two formulations can describe the same feasible set and objective, yet lead a solver through different presolve reductions, relaxation bases, cuts, incumbent searches, branching decisions, and feasibility checks. Generic scaling routines can inspect coefficient magnitudes after export. They usually cannot tell whether a row is a local turbine restriction, a thermal limit, or a system-wide balance. The generator can make that distinction before export. Mainstream modeling environments—Pyomo, JuMP, GAMS, AMPL—are exactly such generators: they assemble the model component by component but, in their standard export paths, hand the solver a flat matrix that no longer marks which rows are local, which variables a component owns, or which rows couple components. Block-structured extensions of these ecosystems do transport such structure—graph-based and block-decomposition modeling layers, and solver-side decomposition files—but they feed it to decomposition algorithms rather than to numerical preprocessing, which is the gap addressed here.

This is not made obsolete by solver-internal scaling. Commercial solvers already contain sophisticated numerical preprocessing, and the experiments below leave it enabled; the question here is complementary—whether an external layer can supply structural numerical information before the solver sees the matrix as an anonymous list of rows and columns. A positive answer would not replace solver scaling, but change the representation on which the solver’s pipeline starts.

The resulting question is simple: can generator-level block information guide positive row scalings that improve the solver-facing representation without changing the MIP? The question is not whether positive row scaling is valid. Multiplying a complete constraint by a strictly positive scalar preserves the feasible set. The question is whether the scaling factors should depend on information that the generator still has: local rows, component blocks, and coupling rows. In this sense, the paper treats pre-export metadata as numerical information rather than only as modeling bookkeeping.

We answer this question with a structure-aware row-scaling framework. The method scales only whole constraints, including their right-hand sides. Hence it preserves the original mixed-integer optimization problem. Its novelty lies in the order and source of the scales, not in claiming that row scaling is new. The generator first scales local component rows. It then summarizes the scale of each locally scaled block. Finally, it scales global coupling rows with the scales of the blocks they connect. Throughout the paper, this local–block–coupling architecture is treated as a *candidate* structure-aware policy whose stages, coverage, and solver dependence are evaluated—and partly refuted—empirically.

The contribution is not a new equilibration kernel—the per-row factor is a known geometric-mean normalization, and the results show that this kernel carries most of the effect. The central contribution is instead a controlled study of *coverage versus selectivity*: whether restricting a known per-row scaling kernel to the rows a generator’s role metadata selects helps or hurts, relative to applying it to every row. Concretely, the paper contributes: (i) a controlled empirical study of that coverage-versus-selectivity trade-off—via a role-blind control run on all three classes—showing that an incomplete structural taxonomy can miss the numerically dominant rows and that the best selection policy depends on the instance class and the solver; (ii) an algebraically exact, auditable row-scaling layer for generated block-structured MIPs, with the returned solutions verified on the original, unscaled constraints; (iii) the use of generator metadata as a *selection policy* that decides which rows to normalize, with idealized conditioning arguments motivating the local–block–coupling order; and (iv) an explicit separation of numerical improvement from computational acceleration, demonstrating that the two need not coincide. Figure 1 summarizes the proposed layer at a glance.

The numerical rationale is also local–block–coupling. In an ideal block-diagonal matrix, scalar block scaling can reduce scale separation between blocks but cannot repair a block’s intrinsic conditioning; once coupling rows are added, their relevant

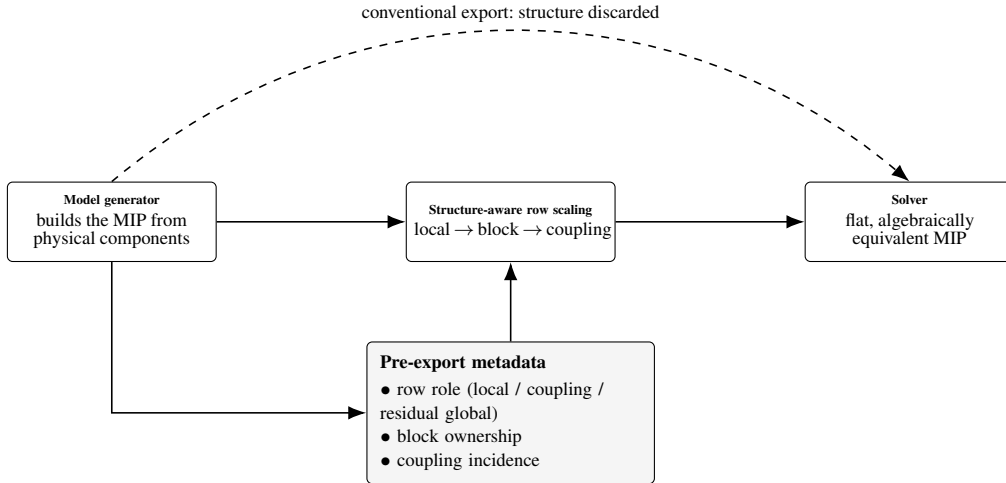


FIGURE 1 The proposed layer in one picture. A generator retains each row’s role—local to a component, coupling between components, or residual global (global but not coupling)—block ownership, and coupling incidence before export. A conventional export (dashed) hands the solver a flat matrix in which this structure is lost. The structure-aware layer instead uses this metadata to assign positive row factors in a local–block–coupling order and exports an *algebraically equivalent* flat MIP; residual-global rows are left unchanged by the selective policy. The solver still receives a single flat matrix; only its numbers change. (The role-blind flat control and the coverage cost of this selectivity are analyzed in [section 6.8](#).)

magnitude is their size relative to the scaled local blocks, not their raw coefficients. Idealized conditioning results make this precise and justify the rule’s architecture, without claiming to predict mixed-integer runtime.

We test the method on short-term hydrothermal dispatch, used not as a new formulation but as a demanding testbed: it creates local hydro, thermal, market, shortage, and demand modules linked by global balances, with coefficients mixing water volumes, turbine flows, power outputs, costs, and penalties—block structure with strong scale heterogeneity. The intended generality, however, is the generated block-structured MIP, not the hydrothermal equations. The layer requires only metadata that many modular generators already have—which rows are local, which block owns a variable or constraint, which global rows couple blocks—while reservoir dynamics, turbine curves, and market costs determine the instances but never enter the scaling rule. The hydrothermal model thus supplies a realistic stress test for a layer that targets any generated MIP with local components and sparse global coupling.

The hydrothermal formulation used in the experiments descends from models for the Río Negro hydroelectric complex and extensions to hydrothermal dispatch with Salto Grande, developed by Risso and coauthors, together with a modular C++ implementation used as the experimental platform^{1,2,3}. We take the formulation as given and insert the proposed preprocessing layer after model generation and before solver invocation.

The paper makes three claims of different strength: the algebraic claim is exact (positive row scaling preserves the MIP); the numerical claim is empirical (structure-aware scaling lowers path-specific solver-facing conditioning diagnostics in the tested instances); and the performance claim is conditional (under intent-to-treat timing, positive row scaling under Gurobi reduces runtime or deterministic-effort metrics in the three classes; after penalizing solutions that fail original-scale verification, the robust runtime advantage remains clearest in SG-Ter-Mer and does not survive in Simple or Full—and under CPLEX 22.1.2.0 there is no robust gain under either reading). A controlled ablation further sharpens the attribution: the conditioning improvement and most of the Gurobi runtime effect flow through the shared per-row normalization kernel, while the metadata acts as a class-dependent selection policy that determines *where* that kernel is applied. The method changed the representation in the intended direction on the operational testbed; acceleration depends on how the solver uses it, and the rule that best conditions the matrix is not the rule that solves fastest.

The rest of the paper follows this chain of evidence. [Section 2](#) identifies the research niche. [Section 3](#) defines the generated model structure and testbed. [Section 4](#) gives the scaling method. [Section 5](#) explains how the experiments test the claims. [Section 6](#) reports numerical and runtime evidence. [Section 7](#) interprets that evidence and answers likely objections. [Section 8](#) states the main limitations and reproducibility implications, and [section 9](#) summarizes the contribution.

2 | RELATED WORK

The literature sets the stage but leaves a specific niche. Hydrothermal-dispatch models provide realistic generated block-structured MIPs with heterogeneous units and global balances; numerical-scaling work explains why coefficient magnitudes matter, but its standard tools operate on the exported matrix; and formulation-quality practice shows that algebraic representation can matter as much as the underlying set. The missing argument is narrower than “scaling is useful”: when a generator still knows each row’s role and each variable’s block, that metadata can be used as numerical information before export. The resulting transformation must then be evaluated separately as an algebraic equivalence, a representation change, and a solver-dependent performance intervention.

2.1 | Hydrothermal dispatch and MIP formulations

Hydrothermal dispatch has been studied with dynamic programming, stochastic dual dynamic programming, nonlinear programming, and mathematical programming. Classical references emphasize the trade-off between thermal generation costs, hydro opportunity costs, demand satisfaction, and intertemporal reservoir management^{4,5}. In short-term settings, MIP formulations are attractive because they incorporate discrete decisions, piecewise-linear approximations, local operating constraints, and system-wide balance equations in a single model⁶.

This literature motivates the experimental domain but not the novelty claim. The dispatch model is used because it is a representative generated block-structured MIP: local physical modules are created separately, their variables have heterogeneous units, and a smaller set of global rows links the modules through system-wide balances. These are precisely the structural features exploited by the preprocessing layer.

2.2 | Dispatch models used in this study

The computational experiments use a reference formulation from a research line on MIP-based dispatch models for the Uruguayan power system. Risso et al. introduced a mixed combinatorial optimization model for the Río Negro hydroelectric complex¹ and extended the modeling line to hydrothermal dispatch for the Uruguay River hydroelectric plant, with emphasis on Salto Grande and coupled hydrothermal operation², and an enriched formulation of the Río Negro model, the closest published reference for the dispatch features used by the Full class⁷. The implementation used here builds on a modular C++ version of the Río Negro dispatch model³.

This provenance fixes the scope of the experiments. The dispatch formulation, data pipeline, and physical components are taken as given. The proposed method acts after the model has been generated and before the solver is called. Consequently, any observed difference between variants is attributed to an algebraically equivalent representation of the same generated MIP, not to a modified dispatch model.

2.3 | Numerical scaling and solver-facing representation

Scaling is a standard preprocessing operation in linear and mixed-integer optimization. Multiplying a constraint by a positive factor preserves the feasible region but changes the numerical magnitudes passed to the solver. These magnitudes can affect feasibility tolerances, relaxation stability, presolve transformations, cut generation, basis quality, and the numerical behavior of LP relaxations. Practical guidelines for difficult LP and MIP models warn that large coefficient ranges and poor scaling can harm solver performance even when the mathematical formulation is correct^{8,9}.

This paper uses that observation in a restricted way: it does not claim coefficient compression is itself an objective, nor that a smaller coefficient range must speed up branch-and-bound. A MIP solver is a pipeline—presolve, internal scaling, root relaxation, cuts, heuristics, incumbents, branching, node processing, termination—whose stages can respond differently to the same external scaling. The experiments therefore distinguish exported coefficient-range ratios from solver-level conditioning diagnostics and from runtime, a distinction central to interpreting the Gurobi–CPLEX contrast.

2.4 | Generic equilibration and its limitation for generated models

Generic equilibration procedures balance rows and columns according to matrix norms. This is a mature subfield of numerical linear algebra. Geometric-mean and arithmetic-mean (ℓ_∞) scaling normalize each row or column by an average of its coefficient magnitudes¹⁰; Curtis–Reid scaling chooses logarithmic diagonal factors that minimize a least-squares measure of coefficient spread¹¹; Sinkhorn–Knopp iteration drives a nonnegative matrix toward a doubly stochastic form by alternating row and column normalization¹²; and Ruiz-type scaling iteratively equilibrates row and column norms through diagonal transformations¹³. These methods are the standard tools and are already embedded in solver presolve and internal scaling¹⁴. Their computational payoff, however, is known to be irregular: the largest empirical study of LP scaling to date found that no scaling method consistently improves—and can even degrade—simplex performance across solvers¹⁵, an early warning that coefficient compression alone is not the lever. Such methods are useful because they are model-agnostic and can be applied after export. That strength is also their limitation in the setting considered here. A flat-matrix method can see coefficient magnitudes and sparsity, but it is not told whether a row is a local turbine restriction, a thermal bound, a market-cost relation, a shortage constraint, or a system-wide balance.

Against this background, the proposed rule is not a new equilibration kernel: its local row factor is, in isolation, a geometric-mean normalization of the row’s nonzero coefficient magnitudes, optionally augmented with the right-hand side (the augmented SA-Aug kernel includes $|b_r|$ among the extremes; the matrix-only SA-Mat kernel does not). What is new is *where the scales come from and in what order*—the factors are keyed to generator metadata (row role, block ownership, coupling incidence) rather than the flat matrix, and applied in a local–block–coupling order so coupling rows are normalized relative to the already-scaled blocks they connect, not their raw coefficients. A flat method (Ruiz, Curtis–Reid, geometric-mean, Sinkhorn–Knopp) cannot make this local-versus-coupling distinction, because role information is exactly what export discards. Concretely, take two rows with identical coefficients: a flat method cannot tell them apart and scales them identically, whereas the structure-aware rule need not. If one is a local block row and the other a coupling row, the local row is scaled by its own coefficients while the coupling row is scaled relative to the *already-scaled* blocks it links—so a coupling row with small raw coefficients can still receive a large factor when it joins a poorly scaled block, and conversely. Identical numbers, different roles, different factors: that is the distinction a flat matrix cannot expose—though, as [section 6.8](#) shows, this representational capability is not itself the runtime lever. The model stays monolithic, as with generic scaling: an external, auditable layer that uses generator metadata before the exported matrix loses it.

2.5 | Formulation quality, generated structure, and decomposition

The MIP literature also stresses that formulation quality matters: two algebraically related formulations can describe the same modeling intention but lead to different relaxations, numerics, and search behavior⁶. The present work is adjacent to that tradition but changes a different object. It does not strengthen the formulation, add valid inequalities, alter variables, or change the feasible set. It changes only the positive row multipliers applied to complete constraints.

The method is also adjacent to structural decomposition. Like Dantzig–Wolfe, Benders, or Lagrangian approaches, it recognizes local blocks and global linking rows. Unlike those methods, it does not decompose the optimization problem, introduce a master problem, or exchange prices or cuts between subproblems. The solver still receives one monolithic MIP. The use of block structure is numerical rather than algorithmic decomposition: block metadata guide the representation presented to the solver.

2.6 | Remaining niche

The resulting contribution is deliberately narrow: it tests whether local rows, block scales, and coupling rows should be treated as different numerical objects before a generated block-structured MIP is exported. The performance comparison rests on distributional evidence over heterogeneous instances rather than a single runtime average, using performance profiles¹⁶, paired tests, deterministic-effort metrics, and solver-level conditioning diagnostics.

3 | MATERIALS AND METHODS: GENERATED MODULAR FORMULATION AND TESTBED

We now define the objects that the scaling rule changes and the objects it must leave unchanged. The generator supplies the block structure; the preprocessing layer changes the row magnitudes; the solver then receives an algebraically equivalent MIP. The dispatch model itself is not new. What matters here is the structure that the generator exposes and the original scale in which all feasibility and objective checks are evaluated after the solve. We consider generated modular MIP instances formulated as

$$\begin{aligned}
 \min_x \quad & c^\top x \\
 \text{s.t.} \quad & Ax \leq b, \\
 & x_j \in \mathbb{Z}, \quad j \in \mathcal{I}, \\
 & x_j \in \mathbb{R}, \quad j \notin \mathcal{I}.
 \end{aligned} \tag{1}$$

In the general generated setting, the vector x collects all variables created by the modules, while A and b contain both local component constraints and global linking constraints. In the hydrothermal testbed used here, x includes generation, reservoir, flow, spillage, shortage, and operating variables. Equalities are represented either explicitly or by pairs of inequalities, depending on the solver interface.

The notation is intentionally application-agnostic: the layer uses the generator’s structural map—row role, block ownership, coupling incidence—not hydrological, thermal, or economic semantics. The hydrothermal variables above explain the physical origin of the coefficients; they are not assumptions of the scaling method.

3.1 | Hydrothermal dispatch testbed and scope

The computational testbed uses the reference model generated by the modular dispatch platform developed around Río Negro and Salto Grande hydrothermal dispatch models^{1,2,3}. The application is useful because it exhibits the target structure: local modules with heterogeneous physical units and global rows that couple those modules through system-wide balances. The hydraulic, thermal, and market-cost constraints are those of the reference model and its implementation.

A caveat about those heterogeneous units is in order, because it is tempting to read the imbalance as artificial. Unit heterogeneity (reservoir volumes in hm^3 , flows in m^3/s , power in MW) lives on the *columns*, and rescaling units is, mathematically, column scaling: an exact reformulation for continuous variables, whereas arbitrary continuous column scaling does not in general preserve the standard integer lattice unless the variable domains are transformed consistently, so a straightforward unit normalization of a MIP is in practice partial. Our column-normalizing baseline (Ruiz+Cols) is the weakest variant in runtime, and the base is no ill-scaled straw man: it already runs under the solver’s own internal equilibration (section 6.7). In the tested configurations, the row-based variants were therefore more effective than the particular column-inclusive baseline considered here; whether a better-designed column or unit normalization could compete is untested, and the row-only scoping decision is revisited as a limitation in section 8.

The method acts on the generated algebraic model, not the physical dispatch formulation: every transformation must preserve the feasible set and let objective values, constraint violations, and operational variables be read in the original scale after solution. It never uses a hydro-specific rule (reservoir priority, turbine efficiency, unit commitment) to choose factors—those details generate the rows, and only their structural role in the block-coupled MIP is used afterward.

3.2 | Block structure

The method targets models generated by components or agents. Each component $i \in \mathcal{B}$ creates local variables x_i , contributes local constraints $A_i x_i \leq b_i$, and appears in a smaller number of global constraints. Let $\mathcal{R}_i$ denote the local row index set of component i . In the hydrothermal testbed, these components include hydro, thermal, market, and demand modules. If u_{loc} stacks

all local variables and g denotes an aggregate coupling variable, the row structure can be written schematically as

$$\begin{bmatrix} D & 0 \\ 0 & I_T \\ L & -I_T \end{bmatrix} \begin{bmatrix} u_{\text{loc}} \\ g \end{bmatrix} \sim \begin{bmatrix} b \\ d \\ 0 \end{bmatrix}, \quad D = \text{diag}(A_1, \dots, A_B). \quad (2)$$

The first block contains local component restrictions. The second block represents an aggregate system restriction. The third block contains coupling equations that link local component decisions with the aggregate system variable. In the dispatch application, these rows correspond to demand and balance relationships. More generally, absorbing the aggregate variable and its defining rows into the coupling block, the coefficient matrix takes the block form

$$A = \begin{bmatrix} A_1 & 0 & \cdots & 0 \\ 0 & A_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_B \\ L_1 & L_2 & \cdots & L_B \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_B \\ h \end{bmatrix}. \quad (3)$$

The matrices A_i encode local restrictions. The matrices L_i encode each block's contribution to the global coupling rows, such as demand balance in the hydrothermal testbed.

Row roles and block incidence are separate metadata fields, and the row taxonomy splits the complete row set $\mathcal{R}_{\text{all}}$ into three pairwise disjoint classes: the local sets $\mathcal{R}_i$, the coupling set $\mathcal{R}_{\text{cpl}}$, and a residual set $\mathcal{R}_{\text{other}}$ of rows that are global but not coupling,

$$\mathcal{R}_{\text{all}} = \left(\bigcup_{i \in \mathcal{B}} \mathcal{R}_i \right) \dot{\cup} \mathcal{R}_{\text{cpl}} \dot{\cup} \mathcal{R}_{\text{other}}, \quad \mathcal{R}_i \cap \mathcal{R}_j = \emptyset \ (i \neq j). \quad (4)$$

A coupling row is incident to several blocks through the variables it involves, but it is not classified as a local row of any of them and does not enter the local-row normalization stage; the implementation enforces this by removing role-classified global and coupling rows from every block row set before scaling. The residual class $\mathcal{R}_{\text{other}}$ (for example, the market shortage segments of the testbed) is a genuine design decision, not an afterthought: a policy must be chosen for it—leave it unscaled, normalize it row by row with the same kernel, or diagnose it and normalize only on a trigger. The structure-aware rules studied here take the most selective policy (leave $\mathcal{R}_{\text{other}}$ unscaled), and the role-blind flat control of [section 6.8](#) realizes the opposite extreme (normalize every row); the operational comparison between them is precisely what measures the cost of this choice, which turns out to be class-dependent ([table 18](#)).

This structure suggests three scaling decisions: scale local rows inside each block; compute representative block scales after local row scaling; and scale global coupling rows with those block scales. The goal is to reduce harmful scale heterogeneity without erasing the modular meaning of the formulation.

3.3 | Algebraic equivalence and spectral diagnostics

We restrict the preprocessing layer to positive row scalings. This restriction makes the transformation auditable: variables keep their original units, and the generated MIP remains the same optimization problem. Let $D_r = \text{diag}(d_1, \dots, d_m)$, with $d_r > 0$ for every row. We replace (A, b) by

$$\tilde{A} = D_r A, \quad \tilde{b} = D_r b. \quad (5)$$

For every vector x ,

$$Ax \leq b \iff D_r Ax \leq D_r b, \quad (6)$$

because each component of the inequality is multiplied by a strictly positive scalar. Thus positive row scaling preserves the feasible set, the integer restrictions, and the objective function. Any difference observed after preprocessing must therefore come from the numerical representation seen by the solver, not from a different mathematical MIP.

The theoretical arguments below use the spectral condition number as a diagnostic of linear-algebraic sensitivity. For a real matrix M , define

$$\kappa_2(M) = \frac{\sigma_{\max}(M)}{\sigma_{\min}^+(M)}, \quad (7)$$

where $\sigma_{\max}(M)$ is the largest singular value and $\sigma_{\min}^+(M)$ is the smallest positive singular value. For rectangular matrices, this ratio is used only as a diagnostic over the nonzero singular spectrum; in the square benchmarks below the minimum is taken over the full spectrum, so that singularity is reported as an infinite condition number. This diagnostic does not characterize the full difficulty of a MIP. A MIP solver also depends on presolve, cuts, branching, primal heuristics, incumbent discovery, feasibility tolerances, and integrality. The role of κ_2 is narrower: it provides an idealized matrix argument explaining why local rows and coupling rows should not be scaled in exactly the same way.

3.4 | Ideal block scaling

The block structure is not only a modeling convenience. It identifies what scalar scaling can and cannot repair. In a purely block-diagonal matrix $D = \text{diag}(A_1, \dots, A_B)$, a single scalar per block can remove differences of scale between blocks, but it cannot improve the intrinsic conditioning of an individual block. The following benchmark makes this statement precise in the ideal square nonsingular case.

Proposition 1 (Ideal scalar scaling of block-diagonal matrices). *Let $D = \text{diag}(A_1, \dots, A_B)$, where every A_i is square and nonsingular. Let $M_i = \sigma_{\max}(A_i)$ and $m_i = \sigma_{\min}(A_i) > 0$. Then*

$$\min_{\alpha_i > 0} \kappa_2(\text{diag}(\alpha_1 I, \dots, \alpha_B I) D) = \max_i \kappa_2(A_i). \quad (8)$$

One minimizer is

$$\alpha_i = \frac{1}{\sqrt{M_i m_i}} = \frac{1}{\sqrt{\sigma_{\max}(A_i) \sigma_{\min}(A_i)}}. \quad (9)$$

Proof. For fixed positive scalars α_i , the scaled matrix is block diagonal:

$$\text{diag}(\alpha_1 I, \dots, \alpha_B I) D = \text{diag}(\alpha_1 A_1, \dots, \alpha_B A_B).$$

The singular values of a block-diagonal matrix are the union of the singular values of its blocks. Therefore, for every i ,

$$\kappa_2(\text{diag}(\alpha_1 A_1, \dots, \alpha_B A_B)) \geq \kappa_2(\alpha_i A_i) = \kappa_2(A_i).$$

Taking the maximum over i gives the lower bound

$$\kappa_2(\text{diag}(\alpha_1 A_1, \dots, \alpha_B A_B)) \geq \max_i \kappa_2(A_i)$$

for every admissible block scaling.

Now choose $\alpha_i = (M_i m_i)^{-1/2}$. Then the largest and smallest singular values of $\alpha_i A_i$ are

$$\begin{aligned} \sigma_{\max}(\alpha_i A_i) &= \sqrt{\frac{M_i}{m_i}} = \sqrt{\kappa_2(A_i)}, \\ \sigma_{\min}(\alpha_i A_i) &= \sqrt{\frac{m_i}{M_i}} = \frac{1}{\sqrt{\kappa_2(A_i)}}. \end{aligned}$$

Because the singular values of the complete block-diagonal matrix are obtained by pooling the singular values of its blocks,

$$\sigma_{\max}(\text{diag}(\alpha_1 A_1, \dots, \alpha_B A_B)) = \sqrt{\max_i \kappa_2(A_i)},$$

and

$$\sigma_{\min}(\text{diag}(\alpha_1 A_1, \dots, \alpha_B A_B)) = \frac{1}{\sqrt{\max_i \kappa_2(A_i)}}.$$

The resulting condition number is $\max_i \kappa_2(A_i)$, matching the lower bound. $\square$

This proposition is not used as an algorithmic recipe. The generated MIPs considered in this paper contain rectangular local matrices, inequalities, bounds, redundant rows, and integer variables; computing singular-value-based block scalings would not be a practical external preprocessing rule for the exported MIP. The proposition instead justifies the architecture of the method. First, poor scale separation between blocks should be treated differently from poor conditioning inside a block. Second,

after local row scaling, each block should contribute a representative scale before global coupling rows are processed. The implemented rules therefore replace the ideal singular-value quantities (M_i, m_i) with coefficient-based proxies that are cheap to compute from the generated model. These propositions are design principles, not performance theorems. They should not be read as predictive guarantees for the implemented MIP preprocessing rule; they motivate the distinction between local rows, block scales, and coupling rows.

3.5 | Effect of global coupling rows

Local block scaling does not exhaust the numerical issue because the generated model is not purely block diagonal. In the hydrothermal dispatch testbed, global rows couple local agent decisions through demand and balance equations. The relevant schematic block is

$$A_{\text{bal}} = \begin{bmatrix} D & 0 \\ L & -I_T \end{bmatrix}, \quad (10)$$

where $D = \text{diag}(A_1, \dots, A_B)$ represents the local blocks and L collects the coefficients by which local decisions enter the global balance. If D is nonsingular, then

$$A_{\text{bal}} = \begin{bmatrix} I & 0 \\ LD^{-1} & -I_T \end{bmatrix} \begin{bmatrix} D & 0 \\ 0 & I_T \end{bmatrix}. \quad (11)$$

This factorization separates local scale from coupling scale. The matrix L by itself is not the relevant object: the coupling felt by the balance row after the local block scales are taken into account is LD^{-1} . Thus a row with moderate entries in L can still be numerically influential if some local block contains weak directions.

The next result makes the dependence on the effective coupling norm explicit.

Proposition 2 (Conditioning of a triangular coupling factor). *Let $R \in \mathbb{R}^{m \times n}$ and define*

$$T_R = \begin{bmatrix} I_n & 0 \\ R & I_m \end{bmatrix}. \quad (12)$$

If $\rho = \|R\|_2$, then

$$\kappa_2(T_R) = \frac{\rho^2 + 2 + \rho\sqrt{\rho^2 + 4}}{2} = \left(\frac{\sqrt{\rho^2 + 4} + \rho}{2} \right)^2. \quad (13)$$

The sign of the identity block is immaterial: since $\begin{bmatrix} I & 0 \\ R & -I \end{bmatrix} = \text{diag}(I, -I) \begin{bmatrix} I & 0 \\ -R & I \end{bmatrix}$ and orthogonal factors preserve singular values, the same formula applies to the factor with $-I_T$ in equation (11).

Proof. Let $R = U\Sigma V^\top$ be a singular value decomposition (SVD) of R , with U and V orthogonal. Orthogonal multiplication does not change singular values, and hence T_R has the same singular values as

$$\begin{bmatrix} V^\top & 0 \\ 0 & U^\top \end{bmatrix} \begin{bmatrix} I_n & 0 \\ R & I_m \end{bmatrix} \begin{bmatrix} V & 0 \\ 0 & U \end{bmatrix} = \begin{bmatrix} I_n & 0 \\ \Sigma & I_m \end{bmatrix}.$$

For each singular value σ of R , the nontrivial part reduces to the two-dimensional block

$$B_\sigma = \begin{bmatrix} 1 & 0 \\ \sigma & 1 \end{bmatrix}.$$

The squared singular values of B_σ are the eigenvalues of

$$B_\sigma^\top B_\sigma = \begin{bmatrix} 1 + \sigma^2 & \sigma \\ \sigma & 1 \end{bmatrix},$$

namely

$$\lambda_\pm(\sigma) = \frac{\sigma^2 + 2 \pm \sigma\sqrt{\sigma^2 + 4}}{2}.$$

Their product is one, and $\lambda_+(\sigma)$ is increasing for $\sigma \geq 0$. Additional directions associated with zero singular values of R contribute singular values equal to one. Therefore the largest squared singular value of T_R is $\lambda_+(\rho)$ and the smallest squared singular value is $1/\lambda_+(\rho)$. Consequently,

$$\begin{aligned}\kappa_2(T_R) &= \lambda_+(\rho) \\ &= \frac{\rho^2 + 2 + \rho\sqrt{\rho^2 + 4}}{2} \\ &= \left(\frac{\sqrt{\rho^2 + 4} + \rho}{2} \right)^2.\end{aligned}$$

□

Applied to (11), the proposition shows that the balance row should not be treated as an ordinary local row. Its numerical effect depends on the effective coupling proxy $\|LD^{-1}\|_2$, not only on the raw magnitude of the entries in L . Moreover, if a common positive factor γ is applied to the coupling row, the scaled block satisfies

$$\begin{bmatrix} D & 0 \\ \gamma L & -\gamma I_T \end{bmatrix} = \begin{bmatrix} I & 0 \\ \gamma LD^{-1} & -I_T \end{bmatrix} \begin{bmatrix} D & 0 \\ 0 & \gamma I_T \end{bmatrix}. \quad (14)$$

Thus γ changes both the effective coupling term and the relative scale of the aggregate variable in the balance equation: pushing γ down improves the triangular factor of equation (14) but degrades its diagonal factor, and vice versa. This is the trade-off behind the proposed coupling treatment—global rows are scaled after local rows, using block-level representative scales, rather than by an isolated row norm—and the augmented rule resolves it explicitly by minimizing the product of the two factors (equation (23) in section 4.2). In the implemented variants, D^{-1} and singular values are replaced by coefficient-based block proxies, yielding a cheap operational approximation to the ideal coupling measure.

3.6 | From ideal diagnostics to implementable proxies

The two propositions establish *which* numerical distinctions a structure-aware rule must preserve, and thereby justify the architecture. They imply three design requirements: local scale imbalance should be reduced before global coupling rows are diagnosed; blocks should be summarized only after their local rows are on a comparable scale; and a coupling row should be scaled relative to the blocks it connects, not only relative to its own raw coefficient norm. The implemented method realizes exactly these distinctions through coefficient-based proxies: they are cheap to compute from the generated model before export, auditable because they depend only on visible row and block metadata, and they avoid expensive or unstable singular-value information from the rectangular, redundant, inequality-based matrices that arise in generated MIPs. The row factor based on the geometric mean of the smallest and largest nonzero magnitudes is a robust coefficient-scale proxy that centers each eligible row without changing variables, bounds, integrality, or the objective.

The coupling rule implements the same logic. In proposition 2, the relevant ideal object is the effective coupling magnitude after local scaling, a term of the form LD^{-1} ; since generated blocks are rectangular, possibly redundant, and not inverted by the preprocessing layer, the block scale s_i used below is a representative coefficient scale for block i computed after local row scaling, and the coupling proxy divides each coupling coefficient by the representative scale of the block owning its column. This is a cheap, auditable surrogate for asking whether the coupling row is large relative to the local numerical scale of the blocks it links.

We state the scope of this argument once and plainly: the propositions describe square, nonsingular blocks under exact SVD scaling, the proxies act on rectangular, inequality-based matrices, and we establish no bound relating the proxy scale to the ideal spectral factor in proposition 1. The propositions therefore motivate the design, and the controlled synthetic suite of section 5.2—where blocks are small, singular values are cheap to compute, and conditioning is measurable by construction—tests it empirically with a bounded outcome: it validates the shared per-row proxy under row-multiplicative imbalance and exercises the common block-relative coupling factor on the oversized branch of its collective design case (section 6.6).

This bridge also explains why the proposed method is row-based. Positive row scaling preserves the original units and domains of all variables; integer and binary variables do not require reinterpretation, and objective coefficients are left in their original economic units. Column scaling can be useful as a generic numerical device, so a Ruiz-plus-column baseline

is included in the experiments. It is not part of the proposed structure-aware layer because the central claim is that row-role and coupling metadata alone already provide useful pre-export numerical information while keeping the transformation algebraically transparent.

4 | NUMERICAL SCALING METHOD

The scaling method keeps the same structural roles visible throughout: local rows, blocks, and coupling rows. Local rows receive row factors first. Blocks then inherit representative scales from their scaled local rows. Coupling rows receive their factors last, using the blocks they connect. This order is the algorithmic expression of the numerical argument in sections 3.4 and 3.5 and of the proxy interpretation in section 3.6.

This section defines the numerical transformations tested in the results. The base variant is the unscaled model generated by the platform. The two proposed variants share the local–block–coupling architecture but are *two different structural rules*: they differ both in whether the right-hand side participates in the scale diagnosis and in how coupling rows are treated (a single grid-selected factor in SA-Aug, one factor per row in SA-Mat), so their head-to-head comparison is not by itself a clean right-hand-side ablation. The clean ablation of the right-hand side is the local-only pair of section 6.8, whose members differ *only* in whether b enters the per-row extremes. The comparison with the Ruiz-kernel baselines tests the choice of per-row kernel under the same structural pipeline; the fully role-blind comparison is carried by the flat control (sections 5.4 and 6.8).

4.1 | Base formulation

The Base variant is passed to the solver without external scaling. Internal solver presolve and internal scaling remain enabled unless stated otherwise. All speedups, objective deviations, and original-scale reliability checks are computed relative to this variant on the same instance, solver, and gap setting (algebraic equivalence itself is proved analytically and needs no per-instance check).

4.2 | Augmented structure-aware scaling

The first proposed variant, denoted SA-Aug, computes row factors from both coefficient magnitudes and right-hand sides. The motivation is that a generated row may have moderate coefficients but a right-hand side in a very different physical or economic scale. Such a row can still lead to a poorly balanced numerical representation.

For each local row r , define

$$\mathcal{S}_r^{A,b} = \{|a_{rj}| : a_{rj} \neq 0\} \cup \{|b_r| : b_r \neq 0\}. \quad (15)$$

If $\mathcal{S}_r^{A,b} \neq \emptyset$, let

$$M_r^{A,b} = \max \mathcal{S}_r^{A,b}, \quad m_r^{A,b} = \min \mathcal{S}_r^{A,b}. \quad (16)$$

The local row factor is

$$\alpha_r^{A,b} = \frac{1}{\sqrt{M_r^{A,b} m_r^{A,b}}}. \quad (17)$$

The scaled row is

$$\alpha_r^{A,b} a_r^\top x \leq \alpha_r^{A,b} b_r. \quad (18)$$

The augmented rule is not only a row-by-row normalization. It is a three-stage structure-aware procedure that first rescales local rows, then summarizes the resulting local block scales, and finally uses those block scales to treat coupling rows separately. After local scaling, let $\tilde{a}_{rj} = d_r a_{rj}$ for local rows, where d_r is the factor *actually applied*— $d_r = \alpha_r^{A,b}$ when the candidate factor $\alpha_r^{A,b} = 1/\sqrt{M_r^{A,b} m_r^{A,b}}$ passes the safeguards of section 4.4 and $d_r = 1$ when it is rejected—so the extremes below are taken on the matrix that is actually exported, not on candidate factors. For each local block i , define

$$M_i^{A,b} = \max_{r \in \mathcal{R}_{i,j}: \tilde{a}_{rj} \neq 0} |\tilde{a}_{rj}|, \quad m_i^{A,b} = \min_{r \in \mathcal{R}_{i,j}: \tilde{a}_{rj} \neq 0} |\tilde{a}_{rj}|. \quad (19)$$

The representative scale of block i is

$$s_i^{A,b} = \sqrt{M_i^{A,b} m_i^{A,b}}. \quad (20)$$

For a coupling row r , let $\beta(j)$ denote the local block that owns column j . Columns owned by no block enter at the arithmetic mean of the valid block scales,

$$s_{\text{free}} = \frac{1}{|\mathcal{B}_+|} \sum_{i \in \mathcal{B}_+} s_i^{A,b}, \quad \mathcal{B}_+ = \{i : 0 < s_i^{A,b} < \infty\}, \quad (21)$$

falling back to $s_{\text{free}} = 1$ if no block has a valid scale, and components that contribute no local rows enter at unit scale. The effective coupling proxy of row r is

$$\hat{\rho}_r^{A,b} = \sqrt{\sum_j \left(\frac{|a_{rj}|}{s_{\beta(j)}^{A,b}} \right)^2}, \quad (22)$$

the magnitude of the row relative to the representative scales of the blocks it connects, and $\rho = \max_r \hat{\rho}_r^{A,b}$ is the worst such magnitude over the coupling rows. SA-Aug then computes a *single candidate* factor $\gamma^{A,b}$ for the coupling block, chosen to minimize the conditioning surrogate that the idealized analysis suggests, and applies it to each coupling row that passes the per-row safeguards of [section 4.4](#):

$$\gamma^{A,b} = \arg \min_{\gamma} \Phi(\gamma), \quad \Phi(\gamma) = \kappa_T(\gamma\rho) \kappa_B(\gamma), \quad \kappa_B(\gamma) = \frac{\max(\bar{M}, \gamma)}{\min(\bar{m}, \gamma)}, \quad (23)$$

where $\kappa_T(\cdot)$ is exactly κ_2 of the triangular coupling factor ([equation \(13\)](#)), κ_B is the diagonal spread once the aggregate variable enters at scale γ , and, writing $\kappa_i = M_i^{A,b}/m_i^{A,b}$ for the post-local-normalization coefficient spread of block i ,

$$\bar{M} = \max_i \sqrt{\kappa_i}, \quad \bar{m} = \min_i \frac{1}{\sqrt{\kappa_i}}. \quad (24)$$

These are the proxy extrema each block *would* present if it were virtually expressed relative to its representative scale $s_i^{A,b}$: dividing block i 's coefficients by $s_i^{A,b} = \sqrt{M_i^{A,b} m_i^{A,b}}$ maps its extremes $M_i^{A,b}, m_i^{A,b}$ to $\sqrt{\kappa_i}, 1/\sqrt{\kappa_i}$. This normalization is virtual—the algorithm does not apply $1/s_i^{A,b}$ to the blocks—and enters only the conditioning surrogate for the diagonal factor $\tilde{D}$; the block scales $s_i^{A,b}$ themselves are used only in the coupling proxy $\hat{\rho}_r$ of [equation \(22\)](#). The minimizer is found on the fixed logarithmic grid $\gamma = 10^{k/10}$, $k = -60, \dots, 60$ (121 log-uniform points spanning $[10^{-6}, 10^6]$). The stage is skipped as a whole when the effective coupling is declared moderate, using the same threshold $\rho \leq 10$ in every experiment (this is a per-instance test—on the operational classes it holds throughout, whereas on the synthetic coupling patterns it fails for the oversized instances, which is precisely where the stage activates)—or when the minimizer falls inside the near-identity band $(1/u, u) = (0.5, 2)$ of the safeguards. After these block-level tests, each coupling row is additionally subject to the per-row coefficient-window and reliability-floor safeguards of [section 4.4](#), which may veto the factor on individual rows; the final transformation is therefore a row-selective diagonal scaling with entries in $\{1, \gamma^{A,b}\}$, not necessarily a uniform scaling of the whole coupling block. Since $\rho = \max_r \hat{\rho}_r$ is taken over all coupling rows, the row that determines $\gamma^{A,b}$ may itself be vetoed by the floor, so $\gamma^{A,b}$ is a *single-pass candidate* rather than the minimizer of the surrogate for the final exported transformation; a fixed-point refinement that re-stabilizes the admissible set is a natural improvement we did not pursue, the stage being inactive on all operational classes. The grid search resolves explicitly the trade-off exposed by [equation \(14\)](#): decreasing γ improves the triangular factor but degrades the diagonal one. In the ideal spectral reading ($\rho = \|LD^{-1}\|_2$ and κ_B from the singular values of $\tilde{D}$), submultiplicativity of κ_2 over that factorization gives κ_2 of the scaled balance block at most $\kappa_T(\gamma\rho) \kappa_B(\gamma) = \Phi(\gamma)$, so minimizing Φ minimizes this ideal-scale bound; on the exported rectangular matrices Φ is evaluated from cheap coefficient proxies as a surrogate, no formal bound to the true κ_2 is claimed ([section 3.6](#)), and when the per-row safeguards veto the factor on some rows the factorization-based surrogate motivates the *candidate* but does not describe the exported, row-selective transformation. The released implementation evaluates these proxies from coefficient summaries of the generated model; the exact evaluation order is documented in the reproducibility repository. This sequence is the main structural feature of the proposed method: local physical scales are used first, block scales are estimated after local normalization, and global coupling rows are scaled only after the blocks they connect have been put on comparable numerical footing.

4.3 | Matrix-only structure-aware scaling

The second proposed variant, denoted SA-Mat, uses the same row–block–coupling architecture but excludes b from the diagnosis. The right-hand side is still multiplied by the selected row factor to preserve equivalence, but it does not influence factor selection.

For each local row r , define the nonzero coefficient extremes

$$M_r^A = \max_{j: a_{rj} \neq 0} |a_{rj}|, \quad m_r^A = \min_{j: a_{rj} \neq 0} |a_{rj}|. \quad (25)$$

If the row has at least one nonzero coefficient, the row factor is

$$\alpha_r^A = \frac{1}{\sqrt{M_r^A m_r^A}}. \quad (26)$$

The scaled row is

$$\alpha_r^A a_r^\top x \leq \alpha_r^A b_r. \quad (27)$$

327 This factor centers the smallest and largest nonzero coefficient magnitudes of the row around a common scale. Unlike Euclidean-
328 norm scaling, it does not directly depend on the number of nonzeros in the row.

Let $\tilde{a}_{rj} = d_r a_{rj}$ after local row scaling, where $d_r = \alpha_r^A$ when the candidate factor passes the safeguards of [section 4.4](#) and $d_r = 1$ otherwise, so the block extremes below are taken on the matrix actually exported, as in SA-Aug. For each local block i , define

$$M_i^A = \max_{r \in \mathcal{R}_i, j: \tilde{a}_{rj} \neq 0} |\tilde{a}_{rj}|, \quad m_i^A = \min_{r \in \mathcal{R}_i, j: \tilde{a}_{rj} \neq 0} |\tilde{a}_{rj}|. \quad (28)$$

The representative block scale is

$$s_i^A = \sqrt{M_i^A m_i^A}. \quad (29)$$

329 No additional multiplier is applied to all rows or columns of block i at this stage. The quantity s_i^A is an auxiliary representative
330 scale: it is used only to normalize the contribution of block i when the coupling-row proxy is computed.

For a coupling row r , with the same ownership map $\beta(\cdot)$ and the same convention for blockless columns as in [section 4.2](#), define

$$\hat{\rho}_r^A = \sqrt{\sum_j \left(\frac{|a_{rj}|}{s_{\beta(j)}^A} \right)^2}. \quad (30)$$

If $\hat{\rho}_r^A > 0$, the coupling-row factor is per row,

$$\gamma_r^A = \frac{1}{\hat{\rho}_r^A}, \quad (31)$$

clipped to $[10^{-8}, 10^8]$ and subject to the safeguards below. When the block extremes of [equation \(28\)](#) are computed, the implementation restricts them to coefficients inside a fixed moderation window $(10^{-6}, 10^6)$, so a single extreme coefficient cannot distort s_i^A . If no post-local-scaling coefficient of block i falls inside that window, block i contributes no representative scale and its columns enter the coupling proxy at the matrix-only mean block scale

$$s_{\text{free}}^A = \frac{1}{|\mathcal{B}_A^+|} \sum_{i \in \mathcal{B}_A^+} s_i^A, \quad \mathcal{B}_A^+ = \{i : 0 < s_i^A < \infty\}, \quad (32)$$

331 with $s_{\text{free}}^A = 1$ when $\mathcal{B}_A^+ = \emptyset$; this is the matrix-only counterpart of [equation \(21\)](#), computed from s_i^A rather than $s_i^{A,b}$.

332 SA-Aug and SA-Mat therefore differ in *two* respects: whether b participates in factor selection, and the granularity and
333 criterion of the coupling treatment (one grid-selected γ for the whole coupling block versus one γ_r per row). A difference
334 between them cannot be attributed to the right-hand side alone; the clean right-hand-side ablation is the local-only pair reported
335 with the stage ablation ([section 6.8](#)), which shares every other design choice.

336 4.4 | Safeguards

337 All applied scaling factors—the local row factors and the coupling-row factors in the augmented or matrix-only variants—are
338 subject to the near-identity and admissible-coefficient-range safeguards, applied independently to each factor. The collective
339 SA-Aug coupling factor is additionally subject to the reliability floor defined at the end of this subsection, and the optional local
340 reliability floor of that same paragraph applies to the augmented local factors. The representative block scales s_i are auxiliary
341 quantities used to compute coupling-row proxies; they are not applied as independent row or column scalings.

4.4.0.1 | Near-identity threshold

A factor is applied only if it departs from one by at least a factor of $u = 2$; factors falling in the open band $(1/u, u) = (0.5, 2)$ are treated as negligible and skipped:

$$\theta \text{ is applied} \iff \left(\theta \leq \frac{1}{u} \right) \text{ or } (\theta \geq u), \quad u = 2. \quad (33)$$

This prevents re-scaling rows or coupling rows that are already at a reasonable scale.

4.4.0.2 | Admissible coefficient range

A factor is applied only if it keeps every affected coefficient within the window $[\varepsilon_{\min}, \varepsilon_{\max}] = [10^{-9}, 10^9]$. Writing $c_{\min}$ and $c_{\max}$ for the smallest and largest magnitudes of the affected coefficients, the factor θ is admitted only if

$$\begin{aligned} \varepsilon_{\min} &\leq c_{\min} \theta, & c_{\max} \theta &\leq \varepsilon_{\max}, \\ [\varepsilon_{\min}, \varepsilon_{\max}] &= [10^{-9}, 10^9]. \end{aligned} \quad (34)$$

Otherwise, the factor is not applied and the corresponding row or coupling row is left at its base scale. The coupling factor is additionally hard-clipped to $\gamma \in [10^{-8}, 10^8]$ prior to this check.

4.4.0.3 | Reliability safeguards

The coefficient window bounds the magnitudes of the scaled coefficients, but it does not by itself bound the effect of a small total row factor d_r on the *original-scale* residual. The exported row is d_r times the original, so *if* the returned point satisfies the exported row to absolute residual at most ε , the corresponding original-scale residual is at most ε/d_r ; a small d_r amplifies it. This is a conditional relation, not a solver-level guarantee: commercial solvers enforce feasibility on internally scaled and presolved representations, so a small ε on the exported row is expected rather than certified, and the floors below control the potential amplification from d_r rather than certify the residual. The *coefficient-only* per-row kernel is structurally safe against this amplification for its coefficients: its factor $\alpha_r = 1/\sqrt{M_r m_r}$ maps the largest coefficient to $\alpha_r M_r = \sqrt{M_r/m_r} \geq 1$, so the largest scaled coefficient is at least one (this centers the extremes geometrically around one; it does *not* preserve the original magnitude—for $M_r/m_r = 10^{12}$ a leading coefficient of 10^{12} becomes 10^3), and consequently, under a fixed residual bound on the exported row, the original residual relative to the row's leading coefficient is not amplified above that bound. The *augmented* kernel does not share this property: because $|b_r|$ enters the extremes, a row whose right-hand side dominates its coefficients receives a tiny d_r , and this is visibly the source of the largest absolute entries in the operational audit, which sit on rows of very large right-hand side (section 6.1). Two reliability safeguards are considered: the collective coupling floor, enabled by default, and an optional local floor evaluated separately.

The *collective coupling floor* guards the single SA-Aug coupling factor, which is chosen to fit the largest coupling rows and has no per-row protection: γ is applied to a coupling row only if the row's largest scaled coefficient magnitude remains at least $\varepsilon_{\text{floor}}^{\text{cpl}} = 10^{-3}$; rows below it keep their base scale. This is enabled by default, and it eliminates the common-factor failure mode on the synthetic suite (section 6.6). The optional *local floor* defines a variant we call SA-Aug-LF, which guards the augmented local factors directly: the local factor is applied only if it also satisfies

$$\alpha_r^{A,b} \geq \varepsilon_{\text{floor}}^{\text{loc}} \quad (\text{in addition to equations (33) and (34)}), \quad (35)$$

so that $d_r \geq \varepsilon_{\text{floor}}^{\text{loc}}$ bounds the conditional original residual by $\varepsilon/\varepsilon_{\text{floor}}^{\text{loc}}$ without referencing ε ; a factor below the floor leaves the row at base scale. SA-Aug-LF is *off* by default, so the main results describe SA-Aug without it, and it is evaluated as a distinct variant with $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-2}$ and a sensitivity sweep in section 6.1. One caveat is specific to the local floor: the condition $\alpha_r \geq \varepsilon_{\text{floor}}^{\text{loc}}$ is not invariant to a positive pre-multiplication of the row. Rescaling row r by $c > 0$ changes $\alpha_r \mapsto \alpha_r/c$, so the same final normalized row can pass the floor under one emitted scale and fail under another. This is acceptable here because the criterion is tied to the generator's emitted scale, which is canonical: the dispatch model emits each row in physical units (power, volume, cost), and the original-scale audit measures the residual against those *same* emitted units. The floor and the audit therefore share one reference—the residual bound ε/α_r is exactly the original-scale quantity the audit checks—so the two remain consistent even though the criterion is not invariant to a hypothetical re-emission of the row. An invariant alternative would key the floor on the resulting row, $\max_j |\alpha_r a_{rj}| \geq \eta$ (as the collective floor already does on the scaled coupling coefficients); we did not adopt it because that quantity does not bound the residual, which depends on the row factor α_r rather than on any

single coefficient. Neither floor is a general guarantee: each bounds the residual relative to a specific quantity (the coupling row’s leading coefficient; the local row’s total factor), and a fuller mixed row-scale floor incorporating $|b_r|$ and the generator’s variable-scale bounds is the natural refinement. We report SA-Aug without the local floor as the primary variant precisely so that its unresolved local-reliability behavior is visible rather than masked.

Because each admitted factor is a strictly positive multiplier applied to a complete constraint—both its coefficients and its right-hand side—the feasible set and the objective value are preserved exactly. The safeguards therefore restrict only the numerical aggressiveness of the transformation, never its equivalence to the base model.

4.4.0.4 | Verification protocol

Algebraic equivalence concerns the mathematical program; it does not by itself guarantee that the solution the solver returns for the scaled formulation satisfies the original constraints within a given absolute tolerance. Every experiment therefore follows a fixed protocol: (i) solve the scaled formulation; (ii) evaluate the returned solution on the original, unscaled model (mapping variables back to original units only where column scaling is used); (iii) check the absolute residuals of every row, bound, and integrality restriction against the thresholds of [section 5.5](#)—the formal pass/fail criterion is absolute; right-hand-side-normalized residuals ([equation \(48\)](#)) are reported descriptively when interpreting exceedances; and (iv) treat a failed check as a numerical-reliability event—in an operational deployment such a run would be repeated with more conservative solver settings or fall back to the base formulation. [Section 6.1](#) reports this verification for all campaigns.

4.5 | Ruiz-type baselines

We compare the proposed variants with two *kernel-swap* baselines that run inside the same structural pipeline. The first, Ruiz, replaces the local per-row kernel with iterative max-norm equilibration applied to the block row sets; the second, Ruiz+Cols, adds continuous-column scaling and physical normalization. Both retain the row classification and the shared coupling stage (which never activates on the operational classes, where $\hat{\rho} \leq 10$ throughout, but does act on the synthetic benchmark; [section 6.6](#)). They are therefore *not* fully role-blind flat-matrix methods: they isolate the choice of *kernel* under the same selection policy, while the fully role-blind selection ablation—the same kernel with no roles at all—is the flat control of [section 6.8](#).

Ruiz is deliberately not a weak foil: max-norm equilibration is a standard, strong scheme, and Ruiz+Cols additionally grants it the column-scaling lever the structure-aware rule forgoes. Together with the flat control these baselines split the comparison into two clean questions—does the geometric-mean kernel beat a Ruiz kernel under the same selection policy, and does the role-based selection beat full coverage under the same kernel; the separate “could the solver have done this itself” control is the internal-scaling baseline of [section 6.7](#). Every variant hands the solver the same monolithic MIP; only the pre-export row factors differ.

The final implementation uses

$$K = 4 \tag{36}$$

Ruiz iterations. Binary and integer variables are not rescaled unless explicitly stated, so that bounds, objective interpretation, and integrality remain unchanged.

4.6 | Algorithmic summary

[Algorithm 1](#) collects the augmented rule end to end.

The matrix-only algorithm follows the same local and block stages with the coefficient-only set $\{|a_{ij}| : a_{ij} \neq 0\}$ and the representative block scales and coupling proxies of [equations \(29\) and \(30\)](#); its coupling stage differs in granularity, assigning each coupling row its own factor $\gamma_r^A = 1/\hat{\rho}_r^A$ ([equation \(31\)](#)) instead of the single grid-selected $\gamma^{A,b}$. SA-Mat is therefore a second structural rule, not a right-hand-side ablation of SA-Aug: the two differ both in whether b enters the diagnosis and in the coupling criterion, and only their local-only reductions ([section 6.8](#)) isolate the right-hand side cleanly.

Algorithm 1 Augmented structure-aware row scaling

Require: Matrix A , right-hand side b , row roles (local/coupling/residual-global), block ownership $\beta(\cdot)$, coupling incidence; optionally useLocalFloor and $\varepsilon_{\text{floor}}^{\text{loc}}$ (default: off, describing the primary SA-Aug variant)

Ensure: Scaled matrix $\tilde{A}$, scaled right-hand side $\tilde{b}$

- 1: Initialize row factors $d_r \leftarrow 1$ $\triangleright$ residual-global rows in $\mathcal{R}_{\text{other}}$ keep $d_r = 1$ throughout
- 2: **for** each local block $i \in \mathcal{B}$ **do**
- 3: **for** each local row $r \in \mathcal{R}_i$ **do**
- 4: Build $S_r^{A,b}$ from nonzero coefficients and nonzero right-hand side
- 5: **if** $S_r^{A,b} \neq \emptyset$ **then**
- 6: Compute $M_r^{A,b}$ and $m_r^{A,b}$ using [equation \(16\)](#)
- 7: Set $\alpha_r^{A,b} \leftarrow 1/\sqrt{M_r^{A,b} m_r^{A,b}}$
- 8: **if** the safeguards admit $\alpha_r^{A,b}$ (and, if useLocalFloor, $\alpha_r^{A,b} \geq \varepsilon_{\text{floor}}^{\text{loc}}$) **then**
- 9: Update $d_r \leftarrow d_r \alpha_r^{A,b}$
- 10: **end if**
- 11: **end if**
- 12: **end for**
- 13: **end for**
- 14: Compute the locally scaled coefficients $\tilde{a}_{rj} = d_r a_{rj}$
- 15: Compute block scales $s_i^{A,b} = \sqrt{M_i^{A,b} m_i^{A,b}}$ using [equation \(20\)](#)
- 16: Compute $\hat{\rho}_r^{A,b}$ for every coupling row using [equation \(22\)](#); set $\rho \leftarrow \max_r \hat{\rho}_r^{A,b}$
- 17: **if** ρ exceeds the coupling threshold **then**
- 18: Set $\gamma^{A,b} \leftarrow \arg \min_{\gamma} \Phi(\gamma)$ over the logarithmic grid $\gamma \in [10^{-6}, 10^6]$ ([equation \(23\)](#))
- 19: Clip $\gamma^{A,b}$ to $[10^{-8}, 10^8]$
- 20: **if** $\gamma^{A,b}$ lies outside the near-identity band ([\(33\)](#)) **then**
- 21: **for** each coupling row r **do**
- 22: **if** $\gamma^{A,b}$ keeps row r inside the coefficient window ([\(34\)](#)) **and** above the reliability floor ([section 4.4](#)) **then**
- 23: Update $d_r \leftarrow d_r \gamma^{A,b}$
- 24: **end if**
- 25: **end for**
- 26: **end if**
- 27: **end if**
- 28: Return $\tilde{A} = \text{diag}(d)A$ and $\tilde{b} = \text{diag}(d)b$

5 | EXPERIMENTAL DESIGN

5.1 | Instance classes and scenarios

The experiments are organized around four reader questions: does scaling preserve the original optimization problem, does it improve the numerical representation, does that change solver behavior, and does the effect survive changes in class, solver, and tolerance? The study uses three instance classes generated by the same modular hydrothermal platform, built from operational instances with available real input data. The pools are balanced by construction: every variant is attempted on every instance of a class (204 Simple, 68 SG-Ter-Mer, 34 Full), so no instance is dropped for one variant and kept for another. Valid-run counts still differ slightly across class–scenario–variant cells because individual runs can fail with an error status (an aborted solve, a missing log, an unavailable conditioning read)—a small fraction, about 1–3% per class in the Gurobi scenarios, recorded per run in the artifacts. Descriptive metrics use all valid runs for a row (so a cell may report, say, 195–199 of 204 Simple instances), whereas speedups, tests, and all base comparisons use matched base–variant pairs in which both runs succeeded, and are therefore unaffected by the unbalanced descriptive counts.

TABLE 1 Three instance classes test the method on progressively different generated hydrothermal models.

Class	Configuration	Main role	Instances
Simple	Río Negro + fixed demand + fixed dispatch	Basic numerical behavior	204
Full	Río Negro nonlinear + Salto Grande + thermal units + market	Largest integrated model	34
SG-Ter-Mer	Salto Grande + thermal units + market	Coupled hydrothermal subset	68

The counts report the maximum operational instance pools with available real input data. Scenario tables may show different n values because each aggregate row uses the valid runs available for that solver–gap–variant combination. Pairwise speedups and tests use matched base–variant pairs only.

TABLE 2 Four solver–tolerance scenarios separate solver dependence from gap–tolerance effects.

Scenario	Solver	Relative MIPGap
1	Gurobi	10^{-2}
2	Gurobi	10^{-3}
3	CPLEX	10^{-2}
4	CPLEX	10^{-3}

Deterministic time is reported when available:
Work units for Gurobi and ticks for CPLEX.

Each measurement is tied to the part of the argument it can support: algebraic equivalence is proved analytically, and the numerical reliability of each returned solution is checked by the original-scale feasibility and integrality tests (with variables mapped back to original units only for the column-scaling baseline); the numerical claim by exported coefficient-range summaries and solver-level conditioning diagnostics; the performance claim by matched base–variant speedups, deterministic effort, performance profiles, and primal–dual progress; and solver dependence by separate Gurobi and CPLEX scenarios at two tolerances (the same distinctions return as claim–evidence–limitation in [table 19](#)).

[Table 1](#) separates the three pools: Simple gives the largest sample for distributional evidence, Full is the smallest and most integrated model, and SG-Ter-Mer isolates a coupled hydrothermal subset—so the averages and paired tests do not come from a single homogeneous population.

The class names are application-specific, but the contrast is structural: whether the rule behaves consistently as the generated MIP moves from a simpler modular model to larger or more strongly coupled ones. The conclusions are thus stated at two levels—empirical evidence from hydrothermal dispatch, methodological object the use of generator metadata in block-structured engineering MIPs.

We evaluate four solver–gap scenarios. The time limit is one hour per solve in every case. Internal solver presolve and internal scaling are left enabled, because the goal is to test whether external preprocessing helps in a realistic solver pipeline. All main experiments were run on ClusterUY, a collaborative high-performance computing platform for scientific computing in Uruguay¹⁷.

The four scenarios in [table 2](#) vary two dimensions separately: the stopping tolerance and the solver pipeline. Scenarios 1–2 test Gurobi at two MIPGap values; scenarios 3–4 repeat the comparison under CPLEX and serve as a solver contrast.

The configuration in [table 3](#) matters because the proposed preprocessing is tested on top of each solver’s default numerical pipeline. The CPLEX conclusions below are therefore statements about this version and ClusterUY environment, not claims about all CPLEX releases or parameter choices.

TABLE 3 The computational environment fixes solver versions, time limits, and default presolve settings for all comparisons.

Item	Configuration
Cluster	ClusterUY, partition <code>ute</code>
Processor	Intel Xeon Gold 6138
Threads	16
Memory limit	12 GB
Time limit	1 hour per solve
Gurobi version	13.0.1 ¹⁸
CPLEX version	22.1.2.0 ¹⁹
Presolve	solver default, enabled

Solver-internal scaling and presolve were left at their default enabled settings unless otherwise stated. CPLEX 22.1.2.0 is the version available in the ClusterUY environment used for the long runs; conclusions involving CPLEX should therefore be interpreted as solver-version and environment specific.

5.2 | Controlled synthetic block-structured benchmark

The hydrothermal testbed evaluates the proposed preprocessing rule on realistic operational generated MIPs. To complement this real-data setting, we define a controlled synthetic benchmark whose purpose is diagnostic rather than physical. The synthetic instances are not intended to model a power system. They isolate the numerical mechanism targeted by the method by allowing local-row scale imbalance and coupling-row scale imbalance to be varied independently.

Each synthetic instance consists of B local blocks. Block i contains continuous activity variables $y_i \in \mathbb{R}_+^{n_y}$ and binary activation variables $z_i \in \{0, 1\}^{n_z}$. A nonnegative slack vector s is added to the global coupling constraints to guarantee feasibility. The generic synthetic MIP is

$$\begin{aligned}
 \min_{y, z, s} \quad & \sum_{i=1}^B c_i^\top y_i + \sum_{i=1}^B q_i^\top z_i + P \mathbf{1}^\top s \\
 \text{s.t.} \quad & R_i y_i + G_i z_i \leq r_i, \quad i = 1, \dots, B, \\
 & 0 \leq y_{ik} \leq U_{ik} z_{ik}, \quad i = 1, \dots, B, \quad k = 1, \dots, n_y, \\
 & \sum_{i=1}^B L_i y_i + \sum_{i=1}^B H_i z_i + s \geq d, \\
 & s \geq 0, \quad z_i \in \{0, 1\}^{n_z}.
 \end{aligned} \tag{37}$$

The first group of constraints defines local block restrictions, the second group links continuous activity variables to binary activation variables, and the third group creates global coupling rows. The penalty P is chosen large enough that the slack variable is used only when the activated block capacity cannot satisfy the synthetic demand.

The unscaled coefficients are generated from bounded random distributions with controlled sparsity. Local matrices R_i and G_i define block-internal resource constraints. Coupling matrices L_i and H_i define the contribution of each block to the global rows. The generator records row metadata

$$\text{row_type} \in \{\text{local}, \text{coupling}\}, \quad \text{block_id} \in \{1, \dots, B\}, \tag{38}$$

and column metadata identifying the block associated with each variable. This metadata is the same information required by the structure-aware scaling rules.

Scale imbalance is introduced after the clean synthetic MIP has been generated. For each eligible row r , a positive row multiplier

$$\theta_r = 10^{\xi_r}, \quad \xi_r \sim \mathcal{U}[-S, S], \tag{39}$$

is drawn and applied to the entire row, including its right-hand side. Therefore, the scaled representation is algebraically equivalent to the clean synthetic MIP. The severity parameter S controls the possible scale range. We use

$$S \in \{0, 3, 6\}, \tag{40}$$

TABLE 4 Controlled synthetic benchmark factors isolate the numerical mechanisms targeted by the scaling rule.

Factor	Levels used	Controlled object	Purpose
Scale pattern	none, local, coupling, coupling-uniform, mixed	Row class affected by artificial scaling	Separate local imbalance from per-row and collective coupling imbalance
Severity S	0, 3, 6	Range of row multipliers 10^{-S} – 10^S	Test mild and severe numerical heterogeneity
Blocks and size	$B = 10$, $n_y = n_z = 8$	Number and size of generated local components	Keep a fixed block structure while varying row-scale imbalance
Rows and seeds	8 local rows per block, 3 coupling rows, 20 seeds	Random coefficients and sparsity patterns	Estimate distributional behavior rather than one instance

The synthetic instances are not physical power-system models. They are controlled block-structured MIPs designed to test whether the preprocessing rule responds to the type of scale imbalance it is designed to exploit. The reported synthetic grid uses the middle size $B = 10$, the five patterns, the three severity levels, and twenty random seeds per pattern–severity cell.

corresponding to no artificial imbalance, moderate imbalance, and strong imbalance.

Five scale patterns are considered:

1. **None:** no artificial row scaling is applied.
2. **Local:** only local block rows are multiplied by factors (39).
3. **Coupling:** only global coupling rows are multiplied by factors (39), one independent draw per row—heterogeneous imbalance *among* the coupling rows.
4. **Coupling-uniform:** all global coupling rows are multiplied by a *single* shared factor $\theta = 10^\xi$, $\xi \sim \mathcal{U}[-S, S]$, one draw per instance—a collective, units-mismatch-style displacement of the coupling rows relative to the blocks, which is the case the block-relative coupling stage is designed for.
5. **Mixed:** both local block rows and global coupling rows are multiplied, independently per row.

The “none” pattern is a negative control: a useful preprocessing rule should not create large artificial changes when the representation is already reasonably scaled. The “local” and two coupling patterns isolate the numerical objects distinguished by the proposed method, and the two coupling patterns separate the collective displacement a single block-relative factor can repair from the per-row heterogeneity it cannot. The “mixed” pattern approximates the combined heterogeneity observed in generated models. These controlled patterns are mechanism checks. They verify that the implementation reacts correctly when the source of imbalance is known by construction, but they are not an operational local-only versus coupling-only ablation on the real hydrothermal instances. In the generated instances the local and coupling row sets are disjoint, as in the method’s row taxonomy (section 4): a coupling row is incident to several blocks through its variables, but it is not a local row of any of them and does not enter the local-row normalization stage.

The reported synthetic runs follow the factor grid summarized in table 4, using the five scaling variants in section 5.3 plus the role-blind flat control introduced in section 6.8, the same original-scale feasibility and objective checks, Gurobi with relative MIPGap 10^{-3} , and a 600 second time limit. They are intentionally small: the benchmark is a controlled mechanism check for numerical conditioning and original-scale numerical reliability, not a performance experiment. Runtime claims are reserved for the operational hydrothermal instances.

5.3 | Scaling variants

The compared variants are: Base, the original formulation; SA-Aug, the augmented structure-aware scaling rule; SA-Mat, the matrix-only structure-aware rule with safeguards; Ruiz, a Ruiz-kernel baseline that swaps the local geometric-mean kernel for max-norm row equilibration ($K = 4$ iterations) inside the same structural pipeline; and Ruiz+Cols, the same with continuous-column scaling and physical normalization. Both Ruiz variants keep the row taxonomy and the shared coupling stage and are therefore *not* fully role-blind; the only fully role-blind baseline is the flat control of section 6.8. Binary and integer variables are not rescaled.

5.4 | Additional controls: internal scaling, sensitivity, and stage ablation

Three further controls sharpen the attribution of the main comparison. They reuse the same matched-pairs design, the same instance classes, and the same original-scale verification, and they leave the five variants above unchanged.

5.4.0.1 | Solver-internal-scaling baseline

To separate “structure-aware external scaling helps” from “any extra scaling pass helps, and the solver could have produced it,” we run the *unscaled* model under the solver’s own scaling controls and compare it, within each scenario, against the structure-aware variants run under default internal scaling. For Gurobi we sweep the scaling flag over its admissible settings (no scaling, equilibration, geometric, and aggressive modes); for CPLEX we sweep its read-scaling parameter (none, default equilibration, aggressive). If a tuned internal setting matches the structure-aware runtime on the same instances, the distinctive claim collapses to generic coefficient compression; if it does not, the gain cannot be attributed to a scaling pass the solver could have produced itself. Presolve remains enabled throughout. Internal scaling is varied only in this control’s solver-specific sweep (including the setting that disables it), while the structure-aware variants always run under the solver’s default internal-scaling configuration, as in the main experiments.

5.4.0.2 | Hyperparameter sensitivity

The constants of the proposed rules are, for SA-Aug: the near-identity band $u = 2$ (equation (33)), the admissible coefficient window $[\varepsilon_{\min}, \varepsilon_{\max}] = [10^{-9}, 10^9]$ (equation (34)), the coupling activation threshold $\hat{\rho} > 10$, the collective coupling reliability floor $\varepsilon_{\text{floor}}^{\text{cpl}} = 10^{-3}$, the γ grid (121 log-uniform points on $[10^{-6}, 10^6]$), and the hard clip $[10^{-8}, 10^8]$; the optional SA-Aug-LF variant adds the local floor $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-2}$ (off by default). SA-Mat shares u , the window, and the clip, and adds the moderation window $(10^{-6}, 10^6)$ on the block extremes. The Ruiz iteration count $K = 4$ is a parameter of the baselines, not of the proposed rules. Three sensitivity analyses are reported. On the Simple class under Gurobi we sweep $u \in \{1.5, 2, 4\}$ and the coefficient window over $\{[10^{-6}, 10^6], [10^{-9}, 10^9], [10^{-12}, 10^{12}]\}$: the runtime conclusions are insensitive (sgm within $[2.01, 2.12]$ s across the sweep, against 3.73 s for the base; Table S1, Supporting Information). The collective coupling floor $\varepsilon_{\text{floor}}^{\text{cpl}}$ is swept over three orders of magnitude on independent synthetic seeds (section 6.6; Table S3), and the local floor $\varepsilon_{\text{floor}}^{\text{loc}}$ over $\{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}\}$ on the failing Simple subset (section 6.1; Table S4). The coupling threshold is swept over $\hat{\rho}_{\text{trigger}} \in \{2, 5, 10, 20, 50\}$ on the synthetic coupling patterns, where the stage is active and the threshold could in principle discriminate: the medians are unchanged, because the coupling proxy is bimodal and essentially never lands in that band (section 6.6). On the operational classes the same proxy sits far from the threshold on both sides ($\hat{\rho} \approx 3 \times 10^{-4}$ in Simple, ≈ 1 in SG-Ter-Mer and Full), so no nearby value changes any operational run and the stage ablation bounds the consequence of the stage as a whole. A separate $K \in \{2, 4, 8\}$ sweep for the Ruiz baselines confirms their irregular matched behavior is not an artifact of the iteration count.

5.4.0.3 | Operational stage ablation

The synthetic benchmark isolates local and coupling imbalance by construction, but the operational comparison evaluates the full local–block–coupling architecture. To attribute the operational effect to its stages, we additionally run two single-stage ablations on the hydrothermal classes—local rows scaled but coupling rows left at base scale, and coupling rows scaled but local rows left at base scale—against the same base. Comparing the full architecture with each ablation identifies how much of the Gurobi effect comes from local row normalization and how much from the coupling-row treatment.

5.5 | Metrics and reporting conventions

The analysis reports three families of metrics. First, numerical feasibility of the returned solution is checked on the original, unscaled constraints. Row-only variants do not transform the variables, so the returned primal vector is evaluated directly; when column scaling is used (the Ruiz+Cols baseline), the variables are first mapped back to their original units. The primary post-solve check is the maximum violation of the original constraints, together with the maximum integrality violation. A run is labeled feasible in original scale if the maximum row violation is at most 10^{-4} and the maximum integer violation is at most 10^{-5} . Recomputing the objective in original units is retained as an auxiliary consistency diagnostic, but it is not the primary feasibility criterion. Objective agreement with the base is measured by

$$\Delta z_{\text{rel}} = \frac{|z_v - z_{\text{base}}|}{\max\{1, |z_{\text{base}}|\}}. \quad (41)$$

This deviation is descriptive only, and it is not by itself evidence of an error in the scaled variant, because the base can itself return an invalid claimed optimum (section 6.1). The correctness reference is therefore the best *verified* incumbent on each instance,

$$z_i^{\text{ver}} = \min \{ z_{iv} : \text{run}(i, v) \text{ passes original-scale verification} \}, \quad \Delta z_{iv}^{\text{ver}} = \frac{|z_{iv} - z_i^{\text{ver}}|}{\max\{1, |z_i^{\text{ver}}|\}}, \quad (42)$$

522 taken within each scenario (solver $\times$ tolerance) over the variants that complete and pass verification; the failure analysis of
 523 section 6.1 uses z_i^{ver} , judging the base against it like any variant, while equation (41) is retained in the audit table as the historical
 524 base-referenced spread with both directions flagged.

Second, numerical behavior is a primary outcome, not only an explanatory variable for speed. We distinguish three quantities. The first is the spectral condition number used in section 3.3 to motivate the rule in an idealized linear-algebraic setting. The second is the exported coefficient-range ratio

$$\rho_\kappa(v) = \frac{(\max |a| / \min |a|)_v}{(\max |a| / \min |a|)_{\text{base}}}, \quad (43)$$

where the extrema are computed over nonzero coefficients in the generated matrix. This is a static proxy and should not be interpreted as the condition number of the MIP. The third is the solver-level ratio

$$\rho_\kappa^{\text{solver}}(v) = \frac{\kappa_{\text{solver}}(v)}{\kappa_{\text{solver}}(\text{base})}, \quad (44)$$

525 where κ_{solver} is the basis condition number reported by the solver on the LP obtained by fixing the integer variables to the
 526 incumbent *each variant found* and re-solving with dual simplex (Gurobi's `fixedModel`), measured after the main MIP solve
 527 so it does not enter the timer. A consequence must be stated plainly: because each variant fixes to its own incumbent, the ratio
 528 (44) can reflect not only the change of representation but also differences in the integer decisions the variants reached; it is a
 529 diagnostic of the solver-facing numerics along each variant's own solution path, not of a single common LP. This is one more
 530 reason the diagnostic is read as a partial, not a definitive, mediator of runtime, and why the primary runtime evidence is the
 531 matched timing rather than the conditioning ratio. For CPLEX we also record tree-level KappaStats (maximum sampled basis
 532 condition number, unstable and ill-conditioned fractions, numerical attention score), which sample bases across the actual tree.
 533 Parsed solver warnings are kept as a descriptive alert, not a ranking criterion. Values below one indicate improvement relative
 534 to the base. The paper thus separates the numerical effect from whether it is converted into faster MIP search.

Third, efficiency is measured by wall-clock time, solver time, total time including preprocessing, deterministic time, node counts, iterations, final gap, and the primal–dual integral. For any time measure T_q , the paired speedup is

$$\text{speedup}_q(v) = \frac{T_{q,\text{base}}}{T_{q,v}}, \quad q \in \{\text{solver, total, det}\}. \quad (45)$$

Values above one indicate that variant v is faster than the base. Because solution times are highly skewed, we report shifted geometric means

$$\text{sgm}_s(\{x_i\}) = \exp\left(\frac{1}{n} \sum_i \log(x_i + s)\right) - s, \quad (46)$$

using $s = 1$ s for times and $s = 10$ for node counts. The primal–dual integral is

$$\text{PI} = \frac{1}{T} \int_0^T \gamma(t) dt, \quad (47)$$

535 where $\gamma(t)$ is the relative optimality gap reported by the solver's informational callback, clamped to $[0, 1]$ and set to one before
 536 the first incumbent; the integral is accumulated by trapezoids over the callback stream and T is the total solve time. Because
 537 callback cadence and gap reporting differ between solvers, the absolute level of PI is comparable within a scenario table but
 538 not across solvers. Timeouts are summarized by PAR10: the arithmetic mean of solver time over the instance pool, with each
 539 run that reaches the limit counted as ten times the time limit.

5.6 | Statistical analysis

The statistical analysis is paired by construction: every variant is compared with the base formulation on the same instance, solver, and MIPGap tolerance. Since MIP solution times are strongly skewed and include censored observations, solver-time comparisons use the Wilcoxon signed-rank test on matched, non-double-timeout pairs. Pairs in which both the base and the variant reach the time limit are treated as censored for this paired test; their global effect is still reflected in PAR10 and in the Dolan–Moré performance profiles. Two conventions of this test must be stated plainly. First, in pairs where exactly one side reaches the limit, that side enters at the capped time: the test is a Wilcoxon on *capped* times, an approximation for censored data. The *sign* of every such pair is nevertheless exact—the side that finished below the limit is faster whatever the censored value—so only the rank magnitudes of those pairs are approximate, and their number is small (zero to four pairs per scenario–class cell for the structure-aware variants, none in SG-Ter-Mer); the reported *p*-values should be read as approximate wherever such pairs exist, and a paired sign test on the observable wins—which is exact under one-sided censoring—agrees in direction with the significant Wilcoxon results in every cell but one (Simple under Gurobi at 0.1% for SA-Mat, where wins sit at parity, 95 against 100, and the Wilcoxon significance comes from the magnitudes of the improvements rather than from their count). Second, as the effect size we report the matched-pairs rank-biserial correlation $r_{rb} = (T^+ - T^-)/(T^+ + T^-)$, computed from the same signed ranks as *W*, together with the median of the per-instance paired speedup ratios; both are genuinely paired quantities, unlike distribution-overlap measures such as Cliff’s δ , which we do not use. We also report the raw *p*-value and the Benjamini–Hochberg adjusted value q_{BH} . The Benjamini–Hochberg correction is applied within each scenario (solver $\times$ tolerance), over the family of all four scaled variants across the three instance classes (twelve paired tests); the tables display the two structure-aware variants, while q_{BH} reflects that full family. In the synthetic benchmark, the adjustment is applied jointly over the 60 paired condition-number comparisons—four variants against the base, five imbalance patterns, three severity levels—within the condition-number metric; Ruiz+Cols is excluded from that family because it coincides exactly with Ruiz on the synthetic instances (no unit metadata), so including it would duplicate a full set of hypotheses. With the convention used here, negative values of r_{rb} indicate that the variant tends to be faster than the base, whereas positive values indicate deterioration.

A word on what the instances are replicates of. The 204, 68, and 34 instances of a class are distinct operational scenarios—different inflow, demand, and cost trajectories and initial conditions—of the *same* physical system and generator, so they are not independent draws from a population of models: they share the component structure, the row taxonomy, and, where scenarios correspond to related operating periods, possibly correlated data. The paired design conditions on the instance, so each test compares variants within a scenario; but to the extent that scenarios are correlated, the effective number of independent replicates is smaller than the nominal *n* and the reported *p*-values may understate uncertainty. We therefore read the tests as evidence about scenarios of this platform, lean on effect sizes and cross-class consistency rather than on any single *p*-value, and flag the generalization beyond this generator as the main limitation (section 8).

To connect numerical and algorithmic effects, we report conditioning ratios by class and variant and compute Spearman correlations between solver speedup and two ratios: the exported coefficient-range ratio ρ_K and the solver-level ratio ρ_K^{solver} . The question is whether the solver-level diagnostic is more associated with speedup than the static proxy across the family of formulations. This is inherently a between-formulation comparison, so the primary inference is an instance-clustered bootstrap: we resample whole instances (all variants of an instance together; 298 distinct instances that have at least one complete conditioning–speedup pair, out of the 306 in the three classes—8 are dropped because the base run errored and yields no speedup; 2000 resamples). It leaves two conclusions standing. First, the pooled solver-level association is negative ($\rho = -0.26$, cluster 95% CI $[-0.32, -0.19]$; $n = 1185$ variant–instance observations, seven short of 298×4 because seven variant runs lack the conditioning diagnostic). Second, the solver-level diagnostic is more associated with speedup than the exported-range proxy ($|\rho_{\text{solver}}| - |\rho_{\text{proxy}}|$ cluster CI $[0.16, 0.31]$, excluding zero). For reference, the naive unclustered statistics—which overstate precision because they treat the four variants of an instance as independent—are a first-order partial rank correlation $\rho_{\text{partial}} \approx -0.25$ controlling for instance size (loaded nonzeros) and a Williams–Steiger $Z \approx 6.1$; we do not rest any conclusion on them. Two limitations remain even after clustering. The association is a between-formulation one: it is carried by the Ruiz-kernel base-lines ($\rho_S \approx -0.33, -0.36$) and is essentially absent *within* the two structure-aware variants ($\rho_S = -0.06$ for SA-Aug, $+0.11$ for SA-Mat; table notes), so the solver diagnostic is not a per-instance predictor of speedup for the proposed rules. And the pooled coefficient still mixes classes; computed within a class the sign is the same but the strength varies (-0.31 Simple, -0.13 SG-Ter-Mer, -0.12 Full at 1%). Either way the association is never strong, which is the substantive point: conditioning is at most a partial mediator (section 7).

6 | COMPUTATIONAL RESULTS

The results follow the hierarchy of claims. We first check whether the solver-facing representation improves. We then ask whether Gurobi converts that improvement into lower runtime or deterministic effort. Finally, we use CPLEX as a contrast to test whether the performance effect is solver-independent.

The order matters because numerical improvement and algorithmic acceleration are related but not identical outcomes—a conditioning table alone would imply too much, a timing table alone too little—so the numerical and algorithmic evidence must be read together. The main pattern is clear but bounded: under Gurobi, the structure-aware variants improve solver-facing conditioning and usually reduce time or deterministic effort; under CPLEX 22.1.2.0, the same external scalings do not produce a robust runtime gain.

6.1 | Original-scale numerical verification across the operational campaigns

Before the per-scenario results, we report the original-scale verification promised in [section 5.5](#) for the four operational campaigns; this is not a secondary audit. It complements the algebraic equivalence result—which is proved exactly by the positivity of the factors and needs no empirical support—by assessing a distinct question: whether the finite-precision solution the solver returns satisfies the original formulation within the stated tolerances. After every completed run, the returned solution is evaluated on the original, unscaled constraints (the row-only variants transform no variables; only Ruiz+Cols requires mapping the variables back to their original units); [table 6](#) pools the four scaling variants per scenario and instance class and reports the share of runs meeting the primary criterion (maximum row violation at most 10^{-4} , maximum integrality violation at most 10^{-5}), together with the median and worst row violation and the objective deviation $\Delta_{z_{\text{rel}}}$ of [equation \(41\)](#). Three facts hold uniformly over the 4,806 variant runs. The integrality criterion is met in every run (worst violation 9.9×10^{-6}). The objective recomputed on the original model from the returned solution matches the solver incumbent to at most 1.0×10^{-6} , so the transformation bookkeeping itself is exact to solver precision. Every exceedance discussed below is therefore a property of the solution returned by the solver, not an artifact of rescaling or of the bookkeeping that undoes it.

Feasibility pass rates are high wherever the solver returns clean solutions: 89% of runs pooled over the Gurobi campaigns and 94% of the CPLEX runs outside the Simple class meet the strict criterion. The visible exception is Simple under CPLEX (32% pass). The unscaled base is the control that locates the cause: it fails the same checker at essentially the same rate (25–26% pass), on essentially the same instances (151 of the 154 base failures at 1% MIPGap recur under SA-Aug), on the same rows (balance equalities with zero right-hand side), and with the same magnitude (median residual near 3×10^{-4}), while Gurobi passes the identical instances at 99.5%. The exceedance is attributable to the solutions CPLEX returns on this class, not to the external scaling, and is consistent with feasibility tolerances being enforced on the solver’s internally scaled and presolved model rather than on the original rows. A fixed absolute threshold also explains the largest worst-case entries: a residual at solver tolerance on a scaled row maps back to ε/α_r on the original row, and the extremes sit on rows of large magnitude. The worst absolute violation in the table (7.9×10^2 , Simple under CPLEX at 0.1%) carries a relative residual of at most 1.6×10^{-6} on a constraint whose right-hand side exceeds 5×10^8 in original units, and the SG-Ter-Mer maximum of 1.1×10^{-1} is below 4.5×10^{-10} relative to its right-hand side. The remaining Gurobi exceedances are a bounded tail: the scaled variants exceed the absolute threshold in 13–15% of Simple runs (22–28% for the Ruiz row variant, 6–15% for the structure-aware variants), and in the worst cell, Full under SA-Aug at 1% (79% pass), the seven failing runs violate original rows by 2.8×10^{-4} to 2.6×10^{-3} (all but one on equality rows) with zero integrality violation and objective agreement with the base within the gap band.

Objective agreement must be read against the MIPGap. The MIPGap bounds each incumbent against its own dual bound, not two incumbents against each other, so “twice the gap” is a descriptive rule of thumb rather than a guarantee—it can be violated when the two runs stop at different bounds, when the objective is near zero, or under numerical trouble. With that caveat, differences of order the gap are compatible with, but not guaranteed by, two independent solves stopped at the prescribed relative gap; the median $\Delta_{z_{\text{rel}}}$ is at or below gap level in every cell, and the firmer reference for correctness remains the best verified incumbent used in the failure analysis below. Because the base itself can return numerically invalid claimed optima, a deviation from the base is not by itself evidence of an error in the scaled variant: in the failure analysis below, the reference on each instance is the best incumbent that passes original-scale verification among all runs of that instance, and the base is judged against that reference like any variant. Deviations beyond twice the gap affect 72 of the 4,806 runs (1.5%), concentrate on a handful of instances, and split into two patterns, both internally consistent in the sense above. First, Ruiz-family runs occasionally claim optimality at grossly suboptimal incumbents: a single Full instance drives the Full-class maxima under both

solvers and both gaps, its row-Ruiz run terminating at gap zero on a suboptimal incumbent against the same base incumbent of 1.1×10^6 ; the magnitude is solver-dependent—original-scale cost 1.9×10^8 under Gurobi ($\Delta z_{\text{rel}} \approx 1.8 \times 10^2$) and 5.9×10^7 under CPLEX ($\Delta z_{\text{rel}} \approx 5.3 \times 10^1$). Second, some unscaled-base runs claim optimality at objectives that verified-feasible scaled runs beat by wide margins: on one Simple instance all four variants reach costs 73% below the base’s claimed optimum, and on one Full instance under CPLEX all four variants agree within 0.5% on an objective the base misses by a factor of six, the value the Gurobi campaigns confirm. The audit flags failures in both directions; the worst worse-than-base deviation of a structure-aware variant among claimed-optimal runs is 3.6%, on one Full instance. Runs that fail to complete are excluded from the matched pairs throughout. For runs that complete but exceed the strict original-scale threshold we report two readings rather than choose one. The *intent-to-treat* analyses of the following subsections retain them as returned, since filtering on an outcome correlated with the treatment would bias the runtime comparison. A *reliability-adjusted* complement penalizes every such run as a computational failure at PAR10 cost, exactly like a timeout (the base column is essentially unchanged by this adjustment: it passes at 97.1–100% in the Gurobi scenarios). Because the strict criterion is absolute and therefore not invariant to the physical units of a row, we also report a third, *mixed* absolute–relative reading, defined precisely as follows. For each row the verifier computes the residual $v_r = (a_r^\top x - b_r)_+$ for $\leq$ rows, $v_r = (b_r - a_r^\top x)_+$ for $\geq$ rows, and $v_r = |a_r^\top x - b_r|$ for equalities, and the *right-hand-side-normalized* residual

$$v_r^{\text{rel}} = \frac{v_r}{1 + |b_r|}. \quad (48)$$

The name is deliberate: the denominator uses $|b_r|$ as a row-magnitude proxy, not the full activity $\sum_j |a_{rj}| |x_j|$ or $|a_r^\top x|$, so it under-normalizes rows whose activity is large while the right-hand side is small—in particular zero-right-hand-side balance equalities, where it reduces to the absolute residual and offers no relief. A fuller normalization by $\max\{1, |b_r|, \sum_j |a_{rj}| |x_j|\}$ is the natural refinement; we keep the right-hand-side form because it is the quantity the verifier already records and because it is the conservative choice (it never declares a zero-right-hand-side violation benign). Bound and integrality checks keep their absolute thresholds. A run passes the mixed reading if every row satisfies $v_r \leq 10^{-4}$ or $v_r^{\text{rel}} \leq \tau_{\text{rel}}$ with $\tau_{\text{rel}} = 10^{-6}$ (the order of the solvers’ feasibility tolerance), so that a tolerance-level residual on a row of order 10^8 is not treated like an operationally meaningful violation. The class separation below does not depend on this convention: sweeping $\tau_{\text{rel}} \in \{10^{-7}, 10^{-6}, 10^{-5}\}$ leaves SG-Ter-Mer at zero failures even at the strictest setting and moves the Simple and Full failure counts by at most a few runs per cell. The full per-variant breakdown—failure counts, median and maximum violations, and PAR10 under all three readings for every scenario, class, and scaled variant—is Table S2 of the Supporting Information. The three readings separate the classes cleanly. The SG-Ter-Mer exceedances are pure absolute-threshold artifacts on rows of large magnitude: under the mixed reading every failure vanishes and the reliability-adjusted PAR10 returns to its intent-to-treat value, so the SG-Ter-Mer advantage survives all three readings (under the strict one, SA-Mat still keeps the lowest PAR10, 558 vs the base’s 573 s at 1% and 559 vs 613 s at 0.1%, with a single penalized run per variant, each worth ~ 535 s of PAR10 on this class size—the metric is deliberately coarse). The Simple and Full exceedances are mostly *not* rescued by the mixed reading (their residuals exceed 10^{-6} in relative terms as well), so those PAR10 advantages hold under intent-to-treat but under neither reliability-adjusted reading (in Simple, 12–30 penalized runs per scaled variant put every variant above the base at both gaps; in Full, SA-Aug moves from 3,308 vs 3,385 s to 9,640 vs 3,385 s at 1%). A reader who weighs reliability strictly should therefore rest the Gurobi runtime case on SG-Ter-Mer.

The SA-Aug exceedances in Simple and Full arise from its augmented *local* factors on rows whose right-hand side dominates their coefficients, not from the coupling stage (inactive here), so the collective coupling floor does not address them; the optional *local* reliability floor of section 4.4 (SA-Aug-LF) does. The exceedances of SA-Mat (which excludes the right-hand side from its factor) and of the Ruiz family (a different kernel, and column transformations in Ruiz+Cols) are not explained by this right-hand-side mechanism and are not separately diagnosed here; their per-variant counts are in Table S2. As an exploratory probe, SA-Aug-LF eliminates the observed failure mode on the twelve Simple instances that fail SA-Aug verification, and a floor sweep there behaves as the residual bound predicts (monotone in $\epsilon_{\text{floor}}^{\text{loc}}$, with 10^{-2} the effective value and a retained solver-time advantage over the base at a higher exported condition number; Table S4, Supporting Information). Because that subset is selected by a post-treatment outcome, this is a demonstration on the failing set, not an unbiased estimate: SA-Aug-LF’s full-class reliability and runtime effect, and its behavior on Full, remain untested, and we treat it as exploratory. We report SA-Aug without the floor as the primary variant so that the unresolved local-reliability behavior stays visible.

Table 5 makes the central reliability conclusion checkable in the body: per class and variant under Gurobi, it reports PAR10 under the three readings on a fixed pool (the instances on which the base completed, so a reference exists; timeouts and completed-but-failed-verification runs are penalized at $10\times$ the time limit, and the few solver aborts are penalized identically rather than dropped). The pattern is uniform across both tolerances. In Simple, the scaled variants are below the base under intent-to-treat but far above it under both reliability-adjusted readings, so the Simple advantage does not survive. In Full the

TABLE 5 PAR10 (s) under the three readings—intent-to-treat, strict absolute, and mixed absolute–relative—for the Gurobi scenarios, on the fixed pool of base-completed instances (pool size n ; timeouts, verification failures, and solver aborts penalized at $10\times$ the time limit). n_{fail} counts completed runs failing the strict criterion. The reliability-adjusted advantage survives only in SG-Ter-Mer (mixed reading); the Simple and Full intent-to-treat advantages do not.

Scenario	Variant	n	PAR10 reading			n_{fail}
			ITT	strict	mixed	
Simple, 1%	Base	197	12	12	12	0
	SA-Aug	197	3	368	368	12
	SA-Mat	197	2	581	550	19
SG-Ter-Mer, 1%	Base	67	573	573	573	0
	SA-Aug	67	562	1100	562	1
	SA-Mat	67	558	1095	558	1
Full, 1%	Base	34	3385	3385	3385	0
	SA-Aug	34	3308	9640	9640	6
	SA-Mat	34	3473	5586	5586	2
Simple, 0.1%	Base	195	54	54	54	0
	SA-Aug	195	109	752	631	21
	SA-Mat	195	48	967	845	30
SG-Ter-Mer, 0.1%	Base	62	613	613	613	0
	SA-Aug	62	603	1184	603	1
	SA-Mat	62	604	1184	604	1
Full, 0.1%	Base	31	5024	5024	5024	0
	SA-Aug	31	4926	7241	7241	2
	SA-Mat	31	5059	8432	6217	3

TABLE 6 Original-scale reliability verification of the four operational campaigns. Each returned solution is evaluated on the original, unscaled constraints (variables are mapped back to original units only for the column-scaling baseline); a run passes if the maximum row violation is at most 10^{-4} and the maximum integrality violation at most 10^{-5} (section 5.5). Row violations are absolute, in original units; Δz_{rel} is equation (41) against the base incumbent on the same instance.

Scenario	Class	n^a	Pass (%)	Base (%) ^b	viol. med.	viol. max.	Δz_{rel} med.	Δz_{rel} max.
Gurobi, 1%	Simple	792	86.7	99.5	2.9×10^{-6}	4.9×10^{-3}	3.8×10^{-5}	8.7×10^{-1}
	Full	134	89.6	97.1	4.8×10^{-6}	2.6×10^{-3}	9.2×10^{-4}	1.8×10^2
	SG-Ter-Mer	270	98.1	100.0	6.8×10^{-7}	1.1×10^{-1}	7.9×10^{-5}	2.1×10^{-2}
Gurobi, 0.1%	Simple	787	85.4	99.5	3.5×10^{-6}	1.2	2.7×10^{-5}	8.7×10^{-1}
	Full	122	90.2	100.0	4.8×10^{-6}	4.5×10^{-3}	6.3×10^{-5}	1.8×10^2
	SG-Ter-Mer	255	98.4	100.0	6.6×10^{-7}	1.1×10^{-1}	1.5×10^{-14}	2.1×10^{-2}
CPLEX, 1%	Simple	816	32.0	24.5	2.3×10^{-4}	4.4×10^{-1}	1.4×10^{-15}	7.3×10^{-2}
	Full	136	80.9	75.8	1.4×10^{-5}	1.3×10^{-3}	1.0×10^{-3}	5.3×10^1
	SG-Ter-Mer	272	100.0	100.0	6.8×10^{-7}	1.4×10^{-5}	4.3×10^{-14}	9.7×10^{-3}
CPLEX, 0.1%	Simple	816	32.6	26.0	2.4×10^{-4}	7.9×10^2	4.3×10^{-15}	1.1×10^{-1}
	Full	134	83.6	84.8	1.3×10^{-5}	6.2×10^{-4}	7.0×10^{-6}	5.3×10^1
	SG-Ter-Mer	272	100.0	100.0	6.9×10^{-7}	1.4×10^{-5}	9.1×10^{-15}	2.0×10^{-3}

^a Completed runs pooled over SA-Aug, SA-Mat, Ruiz, and Ruiz+Cols; runs that failed to complete are excluded, and three runs lacking verification output are counted as non-passing.

^b Pass rate of the unscaled base under the same checker (n per scenario: 195–204 Simple, 31–34 Full, 62–68 SG-Ter-Mer).

same holds. In SG-Ter-Mer the strict reading also penalizes the variants, but the mixed absolute–relative reading—which forgives tolerance-level residuals on large-right-hand-side rows—returns them to their intent-to-treat level below the base, so the SG-Ter-Mer advantage survives all three readings. The per-variant, all-scenario breakdown, including the Ruiz baselines, is Table S2.

TABLE 7 At 1% MIPGap, structure-aware variants improve several descriptive intent-to-treat Gurobi efficiency metrics in Simple and SG-Ter-Mer (the reliability-adjusted readings are in table 5).

Class	Variant	n	sgm T	med. sp. ^{det}	PAR10	PI
Simple	Base	197	4.05	1.00	11.6	0.245
	SA-Aug	197	2.20	1.07	2.6	0.242
	SA-Mat	198	2.03	1.06	2.4	0.241
	Ruiz	198	8.27	0.88	26.7	0.300
	Ruiz+Cols	199	7.52	1.02	27.6	0.233
Full	Base	34	82.93	1.00	3,384.8	0.120
	SA-Aug	34	70.56	1.37	3,307.8	0.075
	SA-Mat	34	77.82	1.15	2487.2	0.055
	Ruiz	33	126.94	1.13	4,566.5	0.263
	Ruiz+Cols	33	75.89	1.32	4,696.3	0.239
SG-Ter-Mer	Base	67	14.14	1.00	572.7	0.664
	SA-Aug	67	8.39	1.32	562.2	0.566
	SA-Mat	67	6.75	1.40	21.0	0.531
	Ruiz	68	14.40	1.00	567.3	0.639
	Ruiz+Cols	68	18.35	0.78	585.1	0.662

Notes. sgm T : shifted geometric mean of solver time in seconds, med. sp.^{det}: median deterministic speedup with respect to the base model. PI: primal–dual integral. Lower is better for sgm T , PAR10, and PI; higher is better for med. sp.^{det}. Bold values mark the best descriptive value within each class and metric. In SG-Ter-Mer the large PAR10 separation of SA-Mat reflects a single near-limit instance it solves within the time budget while Base and SA-Aug time out; the shifted-geometric-mean gain (14.1 $\rightarrow$ 6.8 s) is the more robust evidence for this class, and the PAR10 rescue is not reproduced under the internal-scaling and stage-ablation campaigns (sections 6.7 and 6.8), where the instance times out under all variants.

6.2 | Scenario 1: Gurobi with 1% MIPGap

At the looser Gurobi tolerance, the evidence favors the structure-aware family. SA-Mat gives the lowest shifted geometric solver time in Simple and SG-Ter-Mer, and it improves deterministic effort in all three classes. SA-Aug remains the strongest descriptive rule in Full. The Ruiz-type baselines do not tell the same story: they may change coefficient ranges, but their runtime behavior is irregular.

Table 7 shows the descriptive gains span more than one statistic: SA-Mat leads sgm T and PAR10 in Simple and SG-Ter-Mer and SA-Aug leads Full, with no comparable class-consistent pattern for the Ruiz baselines. The SG-Ter-Mer PAR10 margin, however, rests on a single timeout-rescued instance (see the table notes); for that class the shifted-geometric-mean gain is the robust signal. Figure 2 shows the per-instance speedup distributions, figure 3 contrasts the two conditioning diagnostics against speedup, and figure 4 gives the corresponding Dolan–Moré profiles.

6.3 | Scenario 2: Gurobi with 0.1% MIPGap

The stricter Gurobi tolerance tests whether the pattern survives a harder stopping criterion, and it largely does: SA-Mat again gives the best shifted geometric solver time in Simple and SG-Ter-Mer and SA-Aug is stronger in Full, while the Ruiz-type variants remain unstable. Since the matrix-only rule uses no right-hand-side information, much of the useful structural scale information, in this testbed, is already present in the coefficient matrix.

Table 8 confirms it: the Gurobi evidence does not hinge on the looser stopping criterion, and the distributional and diagnostic views at this tolerance preserve the same pattern as their 1% counterparts (Supporting Information, Figures S1–S3).

Table 9 separates two numerical notions that are easy to conflate. Both structure-aware variants drive the solver-level ratio well below one (SA-Mat brings $\rho_{\kappa}^{\text{solver}}$ to 0.07–0.08 in Full and SG-Ter-Mer while compressing ρ_{κ} much like SA-Aug), whereas Ruiz+Cols may compress the exported matrix while worsening the solver-level diagnostic in Full—so the geometric-mean kernel is not interchangeable with the Ruiz kernel even under the same structural pipeline.

The paired tests in table 10 support the descriptive Gurobi pattern. The signs are favorable for both structure-aware variants, and after Benjamini–Hochberg adjustment both are significant in Simple and SG-Ter-Mer in most Gurobi scenarios. In Full, SA-Aug is not significant at the 1% tolerance ($q_{\text{BH}} = 0.15$) but reaches adjusted significance at 0.1% ($q_{\text{BH}} = 0.039$); given the

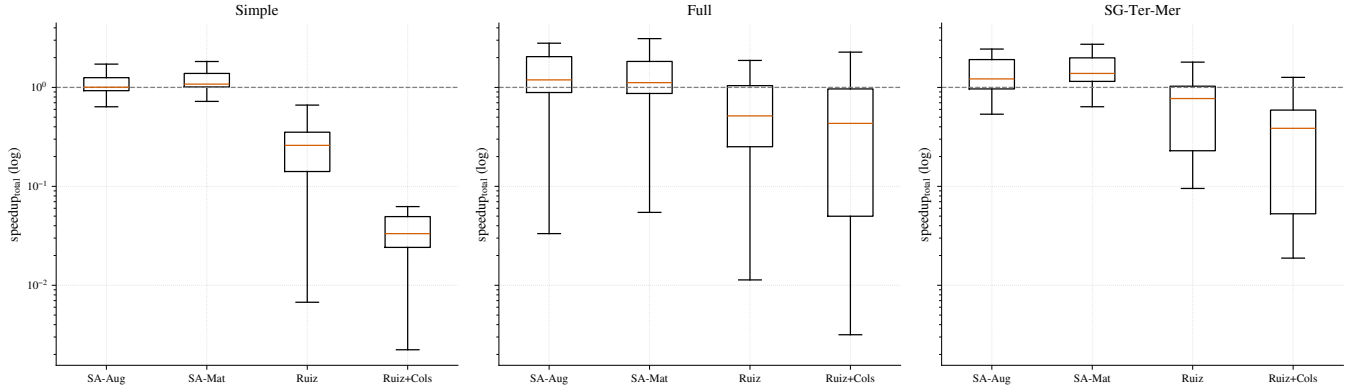


FIGURE 2 At 1% MIPGap, the Gurobi total-time speedup distributions place the structure-aware variants mostly above parity, while Ruiz-type baselines are more dispersed. The axis is the *total*-time speedup (equation (45), $q = \text{total}$), which includes preprocessing; on the easy Simple instances (a few seconds) that overhead dominates and pulls the Ruiz medians well below the solver-time and deterministic-effort parity of table 7 and the $\rho_{\kappa}^{\text{solver}}$ scatter. Values above the parity line indicate acceleration relative to the unscaled model. In all box plots, boxes span the interquartile range (IQR) and whiskers extend to the most extreme observation within $1.5 \times \text{IQR}$; outliers are omitted, since tail behavior is captured by PAR10 and the performance profiles.

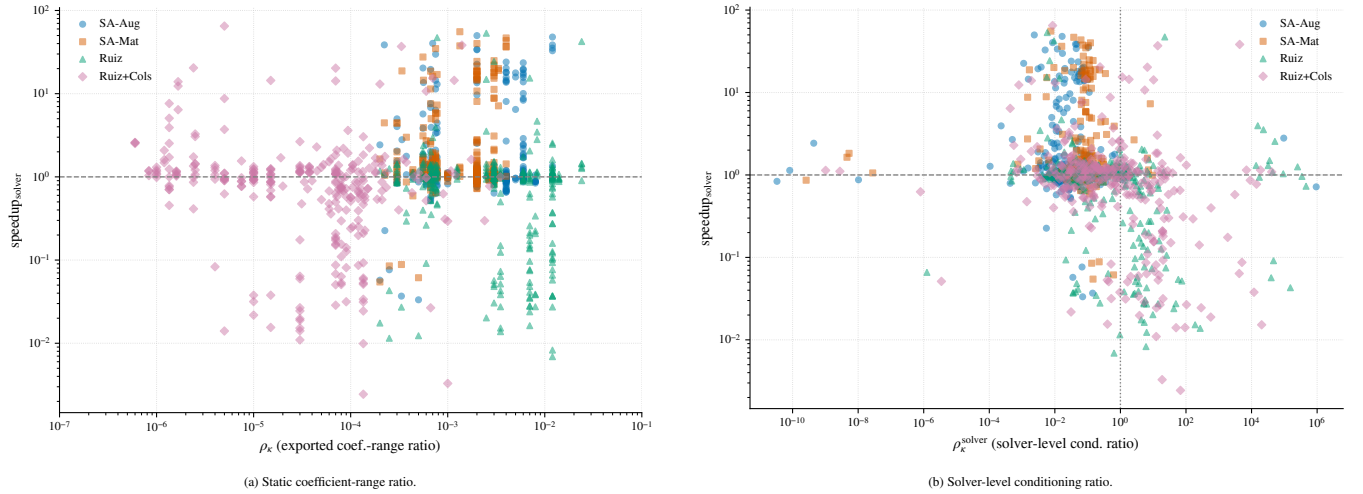


FIGURE 3 At 1% MIPGap, static coefficient-range compression and solver-level conditioning tell different stories. The contrast between panels shows why exported matrix compression alone should not be treated as evidence of easier branch-and-bound search. In panel (a) every static ratio lies below one, so the $\rho_{\kappa} = 1$ reference sits outside the plotted range.

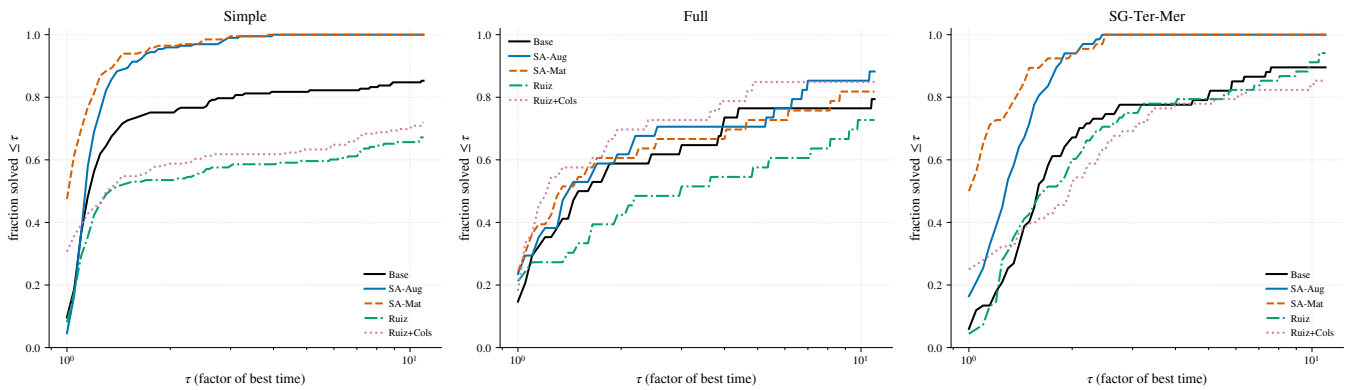


FIGURE 4 At 1% MIPGap, the Dolan–Moré profiles show whether the average Gurobi improvements persist across the instance pool. Curves farther toward the upper-left indicate variants that more often solve close to the best observed time.

TABLE 8 At 0.1% MIPGap, the Gurobi efficiency pattern persists: structure-aware variants remain the strongest structured preprocessing rules.

Class	Variant	n	sgm T	med. sp. ^{det}	PAR10	PI
Simple	Base	195	8.28	1.00	28.5	0.176
	SA-Aug	195	5.65	1.11	28.7	0.180
	SA-Mat	198	5.44	1.10	21.1	0.182
	Ruiz	199	14.12	0.88	41.5	0.261
	Ruiz+Cols	195	12.55	1.02	48.9	0.199
Full	Base	31	305.30	1.00	5,023.8	0.027
	SA-Aug	30	226.90	1.26	3890.2	0.028
	SA-Mat	31	306.39	1.12	5,059.3	0.025
	Ruiz	30	341.56	0.87	5,137.5	0.027
	Ruiz+Cols	31	267.10	1.62	6,140.1	0.022
SG-Ter-Mer	Base	62	14.49	1.00	613.0	0.642
	SA-Aug	63	8.99	1.32	593.7	0.473
	SA-Mat	65	7.97	1.35	22.7	0.439
	Ruiz	65	14.27	1.05	586.8	0.633
	Ruiz+Cols	62	19.16	0.78	629.1	0.625

Notes. Bold values mark the best descriptive value within each class and metric; lower is better for sgm T , PAR10, and PI, whereas higher is better for med. sp.^{det}. In SG-Ter-Mer the PAR10 separation of SA-Mat rests on the same single near-limit rescued instance as at the 1% tolerance (see the notes to [table 7](#)); the sgm gain is the robust signal for this class.

TABLE 9 Structure-aware variants are associated with lower path-specific fixed-incumbent LP conditioning diagnostics under Gurobi at 0.1% MIPGap, whereas static coefficient compression alone is not.

Class	Variant	n	median ρ_κ (IQR)	median $\rho_\kappa^{\text{solver}}$ (IQR)	warnings
Simple	SA-Aug	195	4.0×10^{-3} (4.0×10^{-3})	0.04 (0.08)	1.0
	SA-Mat	195	2.0×10^{-3} (1.0×10^{-3})	0.08 (0.08)	1.0
	Ruiz	192	0.01 (0.01)	0.75 (3.94)	2.0
	Ruiz+Cols	190	1.5×10^{-5} (6.7×10^{-5})	1.19 (12.70)	1.0
Full	SA-Aug	30	5.6×10^{-4} (3.7×10^{-4})	0.02 (0.04)	1.0
	SA-Mat	31	5.6×10^{-4} (3.7×10^{-4})	0.07 (0.08)	1.0
	Ruiz	29	5.6×10^{-4} (3.7×10^{-4})	0.47 (2.55)	2.0
	Ruiz+Cols	30	6.0×10^{-4} (8.0×10^{-4})	1.60 (24.82)	0.0
SG-Ter-Mer	SA-Aug	62	6.7×10^{-4} (1.8×10^{-4})	0.01 (0.01)	1.0
	SA-Mat	61	6.7×10^{-4} (1.8×10^{-4})	0.08 (0.02)	1.0
	Ruiz	61	6.7×10^{-4} (1.8×10^{-4})	0.06 (0.06)	1.0
	Ruiz+Cols	59	1.0×10^{-4} (8.9×10^{-5})	0.07 (0.11)	0.0

Notes. Values below one indicate improvement relative to the base on the same scenario. ρ_κ is the exported coefficient-range proxy, whereas $\rho_\kappa^{\text{solver}}$ is computed from the solver-level conditioning diagnostic on the fixed-integer LP. Because each variant fixes its own incumbent, these ratios describe the numerics along each variant's solution path and do not compare one common LP ([section 5.5](#)). All ratios are strictly positive; the smallest are shown in scientific notation. The warnings column summarizes solver numerical warning flags parsed from logs and is used only as a descriptive diagnostic.

small sample, censoring, and run-to-run variability, both Full results should still be interpreted cautiously. The paired effect sizes range from small or moderate to large (r_{ib} from -0.23 to -0.90), with the strongest paired shifts appearing for SA-Mat in SG-Ter-Mer; the evidence is a distributional shift, not a guarantee that every instance accelerates.

[Table 11](#) adds the aggregate correlation check. Across the pooled family of formulations, the path-specific fixed-incumbent LP diagnostic is more strongly associated with speedup than the exported-range proxy (which is essentially uninformative); the instance-clustered bootstrap confirms this contrast survives the within-instance dependence ([section 5.5](#)). The association is weak, driven mainly by the Ruiz-kernel variants, and provides no useful per-instance runtime prediction for SA-Aug or SA-Mat (essentially null within either; see the table notes). It is enough to prefer solver-facing diagnostics over exported-matrix summaries when reading representation, and the near-null structure-aware correlations reinforce the partial-mediator reading developed in [section 7](#).

TABLE 10 Paired Gurobi tests show a distributional runtime shift in favor of the structure-aware variants, especially in Simple and SG-Ter-Mer.

Scenario	Class	Variant	W	p	q_{BH}	r_{tb}
1% MIPGap	Simple	SA-Aug	7535.0	0.0057	0.0097	−0.227
1% MIPGap	Simple	SA-Mat	3011.0	$< 10^{-4}$	$< 10^{-4}$	−0.691
1% MIPGap	Full	SA-Aug	175.0	0.0984	0.1476	−0.337
1% MIPGap	Full	SA-Mat	172.0	0.1406	0.1875	−0.306
1% MIPGap	SG-Ter-Mer	SA-Aug	501.0	0.0001	0.0005	−0.547
1% MIPGap	SG-Ter-Mer	SA-Mat	115.0	$< 10^{-4}$	$< 10^{-4}$	−0.896
0.1% MIPGap	Simple	SA-Aug	6963.0	0.0010	0.0024	−0.264
0.1% MIPGap	Simple	SA-Mat	7069.0	0.0016	0.0028	−0.260
0.1% MIPGap	Full	SA-Aug	97.0	0.0262	0.0393	−0.487
0.1% MIPGap	Full	SA-Mat	173.0	0.7140	0.7140	−0.085
0.1% MIPGap	SG-Ter-Mer	SA-Aug	456.0	0.0004	0.0018	−0.518
0.1% MIPGap	SG-Ter-Mer	SA-Mat	341.0	$< 10^{-4}$	0.0002	−0.639

Notes. Tests are paired against the base formulation on matched instances. r_{tb} is the matched-pairs rank-biserial correlation computed from the same signed ranks as W ; negative values mean the variant tends to be faster. Double-timeout pairs are excluded from the Wilcoxon test and retained through PAR10 and the performance profiles. The effective number of matched pairs equals the full class pool (Simple 195–197, SG-Ter-Mer 61–67) except in Full, where double-timeouts reduce it to 27–32; zero-difference pairs are dropped by the Wilcoxon zero method.

TABLE 11 Across the pooled scaling family, the path-specific fixed-incumbent LP diagnostic is more associated with Gurobi speedup than the exported-range proxy; within the structure-aware variants the association is essentially null.

Scenario	$\rho_S(\rho_\kappa, sp)$	$\rho_S(\rho_\kappa^{\text{solver}}, sp)$	solver 95% CI	$(\rho^{\text{sol}} - \rho^{\text{prox}})$ CI
1% MIPGap	+0.002	−0.256	[−0.32, −0.19]	[0.16, 0.30]
0.1% MIPGap	+0.010	−0.267	[−0.34, −0.20]	[0.16, 0.32]

Notes. sp denotes solver speedup; ρ_S is the pooled Spearman correlation over all four scaled variants. The confidence intervals are 95% instance-clustered bootstrap intervals (2000 resamples of whole instances, 298 clusters at 1%, 288 at 0.1%)—the primary inference, since each instance contributes four correlated rows: the solver-level association is negative and its magnitude exceeds the proxy’s (the difference CI excludes zero). A naive Williams–Steiger comparison that treats the rows as independent gives $Z \approx 6.1$ (1%) and 6.6 (0.1%); it overstates precision and is not relied upon. Decomposed by variant, the negative solver-level association is carried by the Ruiz baselines ($\rho_S \approx -0.33$ for Ruiz and -0.36 for Ruiz+Cols at 1%), whereas for the structure-aware variants it is essentially null (SA-Aug -0.06 ; SA-Mat $+0.11$), with the same pattern at 0.1%, so it provides no per-instance runtime prediction for the proposed rules.

Together with tables 9 and 11, these results support a two-step interpretation: structure-aware scaling changes the numerical representation seen by Gurobi, and—across the pooled family of formulations, and primarily because of the Ruiz-kernel variants—the path-specific fixed-incumbent LP diagnostic is more associated with speedup than the exported-range proxy. This association is not present strongly enough within SA-Aug or SA-Mat to serve as a per-instance runtime predictor for the proposed rules.

6.4 | Scenario 3: CPLEX with 1% MIPGap

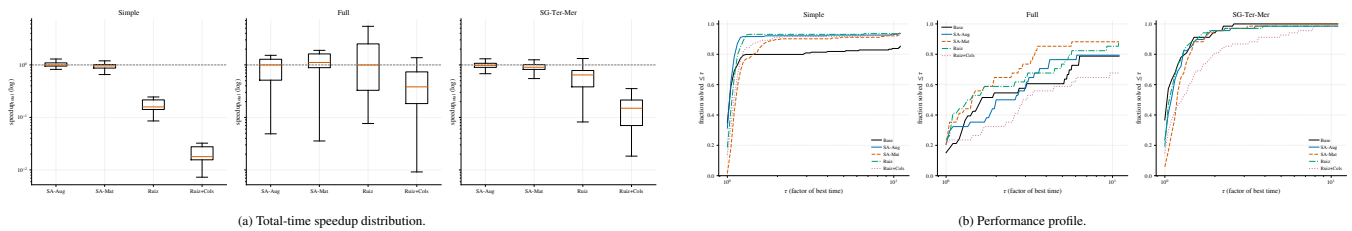
The CPLEX scenarios provide a negative solver contrast to the Gurobi results. The structured variants remain close to parity and do not reproduce the systematic deterministic gains observed under Gurobi. Some descriptive wall-time improvements appear, especially in Full for SA-Mat, but the paired tests do not support robust structure-aware acceleration. In fact, some CPLEX signs are unfavorable for the structured variants. The correct interpretation is therefore not that CPLEX improves less than Gurobi, but that the present external scaling layer does not reliably improve CPLEX runtime in this environment. Ruiz is sometimes competitive in Full, but its performance is not stable across classes.

For CPLEX, table 12 has to be interpreted differently from the Gurobi tables: the class-consistent structure-aware pattern is absent, and in SG-Ter-Mer the base scale already gives the best shifted geometric time and PAR10. The Simple class shows the descriptive–paired dissociation most clearly: SA-Mat lowers the class sgm ($2.23 \rightarrow 1.68$ s) and PAR10 ($6.2 \rightarrow 3.4$ s), yet its paired test is significantly *unfavorable* ($r_{tb} = +0.29$, with 61 paired wins against 143 losses, table 14): a few hard instances in the tail accelerate enough to move the aggregates while the typical instance slows slightly. Where a CPLEX aggregate improves, it is therefore a tail effect, not a per-instance speedup. Figure 5 shows the corresponding distributional and profile views.

TABLE 12 At 1% MIPGap, CPLEX does not reproduce the Gurobi structure-aware acceleration pattern.

Class	Variant	n	sgm T	med. sp. ^{det}	PAR10	PI
Simple	Base	204	2.23	1.00	6.2	0
	SA-Aug	204	1.46	1.04	3.1	0
	SA-Mat	204	1.68	1.03	3.4	0
	Ruiz	204	1.49	0.99	3.6	0
	Ruiz+Cols	204	1.51	0.96	2.6	0
Full	Base	33	158.38	1.00	5621.7	3.1×10^{-8}
	SA-Aug	34	165.43	0.98	5546.0	3.2×10^{-8}
	SA-Mat	34	115.60	1.00	6497.8	2.3×10^{-8}
	Ruiz	34	115.96	1.14	3506.3	2.1×10^{-8}
	Ruiz+Cols	34	225.94	0.70	4681.1	5.5×10^{-8}
SG-Ter-Mer	Base	68	2.21	1.00	3.5	1.0×10^{-10}
	SA-Aug	68	2.37	1.02	9.9	1.0×10^{-10}
	SA-Mat	68	2.41	1.00	5.5	1.1×10^{-10}
	Ruiz	68	2.33	1.01	10.3	9.7×10^{-11}
	Ruiz+Cols	68	3.18	0.88	15.2	1.0×10^{-10}

Notes. Bold values mark the best descriptive value within each class and metric. Since the CPLEX statistical tests do not support a robust positive external-scaling effect, these bold entries are descriptive only. PI is accumulated from CPLEX's own callback stream and is comparable within this table but not against the Gurobi tables (section 5.5).

**FIGURE 5** At 1% MIPGap, the CPLEX distributional and performance-profile views break the Gurobi pattern. The plots therefore serve as a solver contrast.

6.5 | Scenario 4: CPLEX with 0.1% MIPGap

The stricter CPLEX setting confirms the same solver dependence and makes the negative runtime conclusion clearer. The structured variants remain at or near parity in deterministic effort, and the clearest descriptive improvement in shifted geometric solver time comes from Ruiz rather than from the proposed structure-aware rules; the matrix-only rule does lower the shifted geometric time on Full (from 421.6 to 370.5 s), but this is descriptive only, as the paired test on that class is not significant ($p = 0.21$). This is a useful negative result: external scaling may improve some numerical diagnostics without becoming a robust runtime accelerator for CPLEX in this environment.

Table 13 reports the cells; the corresponding distributional and profile views (Figure S4, Supporting Information) remain mixed under the stricter tolerance, supporting the solver-dependent interpretation.

The paired CPLEX tests in table 14 keep the descriptive tables in perspective. Unlike the Gurobi evidence, the signs are small, mixed, and sometimes unfavorable; after multiplicity adjustment they do not support a robust positive runtime effect for the structure-aware variants.

The CPLEX runs nevertheless carry their own conditioning evidence, recorded as tree-level KappaStats (section 5.5), and it moves in the same direction as the Gurobi diagnostics. The median sampled-basis condition ratio of the structure-aware variants relative to the base is below one in Simple (0.84–0.87 at 1%, 0.80–0.81 at 0.1%) and drops by four to five orders of magnitude in Full ($\approx 5 \times 10^{-5}$ at 1%, $\approx 2\text{--}3 \times 10^{-6}$ at 0.1%), while CPLEX's numerical attention score on Full falls from 0.0038 for the base to zero and the sampled unstable fraction vanishes; SG-Ter-Mer sits near parity (0.90–0.99). The numerical claim thus holds under both solvers, even though CPLEX does not convert the improved representation into runtime.

Summarizing the four scenarios by their best descriptive variant makes the solver contrast visible at a glance: the structure-aware rules hold the lowest shifted geometric times in every Gurobi cell (SA-Mat in Simple and SG-Ter-Mer, SA-Aug in

TABLE 13 At 0.1% MIPGap, CPLEX remains close to parity for the structure-aware variants, confirming solver dependence.

Class	Variant	n	sgm T	med. sp. ^{det}	PAR10	PI
Simple	Base	204	5.14	1.00	213.6	1.2×10^{-12}
	SA-Aug	204	4.20	1.02	188.4	6.5×10^{-13}
	SA-Mat	204	4.48	1.01	202.5	5.6×10^{-13}
	Ruiz	204	4.13	1.00	192.0	1.0×10^{-15}
	Ruiz+Cols	204	5.28	0.94	546.6	7.7×10^{-13}
Full	Base	33	421.59	1.00	9162.0	2.7×10^{-8}
	SA-Aug	34	422.62	1.07	8930.4	2.7×10^{-8}
	SA-Mat	33	370.45	1.05	11186.3	2.3×10^{-8}
	Ruiz	34	285.10	1.28	5857.3	1.9×10^{-8}
	Ruiz+Cols	33	377.41	0.96	1.1×10^4	5.2×10^{-8}
SG-Ter-Mer	Base	68	2.82	1.00	18.8	1.1×10^{-10}
	SA-Aug	68	2.84	1.01	20.6	1.0×10^{-10}
	SA-Mat	68	2.84	1.00	15.8	1.1×10^{-10}
	Ruiz	68	2.61	1.02	7.5	1.1×10^{-10}
	Ruiz+Cols	68	4.02	0.85	539.5	1.3×10^{-10}

Notes. Bold values mark the best descriptive value within each class and metric. Under CPLEX, these descriptive minima do not imply a robust positive external-scaling effect.

TABLE 14 Paired CPLEX tests do not support a robust positive runtime effect for the structure-aware variants.

Scenario	Class	Variant	W	p	q_{BH}	r_{tb}
1% MIPGap	Simple	SA-Aug	9377.0	0.2016	0.3456	−0.103
1% MIPGap	Simple	SA-Mat	7460.0	0.0004	0.0023	+0.286
1% MIPGap	Full	SA-Aug	172.0	0.3357	0.4284	+0.209
1% MIPGap	Full	SA-Mat	131.0	0.1042	0.2083	−0.355
1% MIPGap	SG-Ter-Mer	SA-Aug	1068.0	0.5211	0.5211	+0.090
1% MIPGap	SG-Ter-Mer	SA-Mat	740.0	0.0082	0.0326	+0.369
0.1% MIPGap	Simple	SA-Aug	8765.0	0.0453	0.1213	−0.162
0.1% MIPGap	Simple	SA-Mat	10056.0	0.7230	0.7230	−0.029
0.1% MIPGap	Full	SA-Aug	173.0	0.7140	0.7230	−0.085
0.1% MIPGap	Full	SA-Mat	125.0	0.2079	0.3563	−0.288
0.1% MIPGap	SG-Ter-Mer	SA-Aug	1085.0	0.5908	0.7089	+0.075
0.1% MIPGap	SG-Ter-Mer	SA-Mat	985.0	0.2507	0.3760	−0.160

Notes. Tests are paired against the base formulation on matched instances. r_{tb} is the matched-pairs rank-biserial correlation from the same signed ranks as W (negative = variant faster). Under CPLEX, the effect signs are small, inconsistent, or unfavorable. They do not support a robust positive structure-aware runtime effect. The effective number of matched pairs equals the full class pool (Simple 203–204, SG-Ter-Mer 68) except in Full, where double-timeouts reduce it to 26–29; zero-difference pairs are dropped by the Wilcoxon zero method.

Full, at both tolerances), whereas the CPLEX winners are heterogeneous—varying from SA-Aug to Base to Ruiz by class and tolerance—and must be read together with the paired Wilcoxon tests, which do not support a robust positive structure-aware effect there.

6.6 | Controlled synthetic mechanism check

The operational experiments above test the full solver–model interaction. The synthetic benchmark has a different role (figure 6): it checks the numerical mechanism under controlled row-scale imbalance. The grid uses $B = 10$ blocks, each with 8 continuous and 8 binary variables, 8 local constraints plus binary-activation rows, and 3 global coupling rows with feasibility slacks. Row imbalance is injected through the pattern $\{none, local, coupling, coupling-uniform, mixed\}$ and $S \in \{0, 3, 6\}$: in *coupling* each selected row r draws its own multiplier $\theta_r = 10^{\xi_r}$, $\xi_r \sim \mathcal{U}[-S, S]$, whereas in *coupling-uniform* all coupling rows share a single draw per instance (section 5.2). The grid contains 20 seeds per pattern–severity cell, for 300 instances and 1800 variant runs (the five variants of section 5.3 plus the role-blind flat control of section 6.8).

TABLE 15 Controlled synthetic benchmark: median condition numbers before scaling and after each variant, with worst original-scale violations across the selected cells.

Pattern	S	κ^{pre} (Base)	κ^{post} (SA-Aug)	κ^{post} (SA-Mat)	κ^{post} (Ruiz)	κ^{post} (Flat)	v_{row}	v_{int}
none	0	1.34×10^3	1.26×10^4	2.49×10^3	1.19×10^4	1.09×10^3	2.27×10^{-13}	0
local	3	1.84×10^8	1.44×10^4	4.68×10^3	3.51×10^4	1.07×10^3	2.91×10^{-11}	0
local	6	1.49×10^{14}	1.38×10^4	4.02×10^3	3.77×10^4	1.07×10^3	1.49×10^{-8}	0
coupling	3	6.86×10^5	3.47×10^4	2.53×10^3	6.06×10^4	1.22×10^3	1.16×10^{-10}	0
coupling	6	4.87×10^8	3.33×10^5	2.49×10^3	3.58×10^5	1.22×10^3	1.19×10^{-7}	2.03×10^{-6}
c.-unif., $\theta > 1$ ($n=7$)	3	3.16×10^5	3.99×10^3	2.57×10^3	4.52×10^3	1.05×10^3	1.16×10^{-10}	0
c.-unif., $\theta \leq 1$ ($n=13$)	3	4.11×10^4	4.82×10^3	2.51×10^3	4.46×10^3	1.30×10^3		
c.-unif., $\theta > 1$ ($n=7$)	6	6.94×10^7	3.78×10^3	2.57×10^3	4.37×10^3	1.05×10^3	2.38×10^{-7}	3.27×10^{-7}
c.-unif., $\theta \leq 1$ ($n=13$)	6	3.19×10^6	2.36×10^5	2.50×10^3	2.68×10^5	1.29×10^3		
mixed	3	2.06×10^8	5.00×10^4	4.68×10^3	4.86×10^4	1.18×10^3	5.82×10^{-11}	0
mixed	6	1.52×10^{14}	6.01×10^6	4.02×10^3	6.01×10^6	1.22×10^3	1.31×10^{-6}	1.28×10^{-7}

Notes. Values are medians by pattern, severity, and variant, except for the two violation columns, which report worst original-scale violations in the corresponding pattern–severity cell (the coupling-uniform cells are reported whole). The grid contains 20 seeds per cell; the coupling-uniform rows are split by the direction of the shared factor θ , with subgroup sizes in the row labels. Dispersion for the key contrast at $S = 6$: SA-Aug spans $[3.4, 5.2] \times 10^3$ (interquartile) on the oversized branch it repairs, and $[3.8 \times 10^3, 5.2 \times 10^6]$ on the undersized branch its one-sided trigger leaves untreated, whose pre-scaling spread is itself $[3.8 \times 10^4, 6.9 \times 10^7]$. The synthetic benchmark is a conditioning and reliability check, not a runtime benchmark. Ruiz+Cols is omitted as a column: the synthetic instances carry no unit metadata, so its physical-normalization step is inert and it coincides with Ruiz everywhere. All 1800 runs pass the original-scale verification with the reliability floor of section 4.4 enabled (its ablation is analyzed in the text).

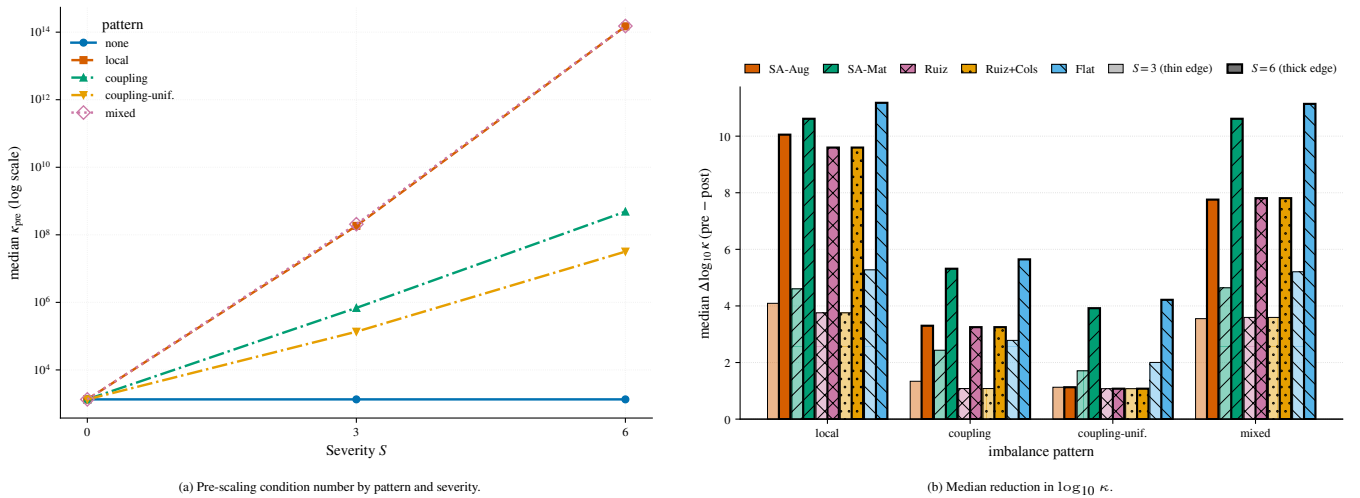


FIGURE 6 Controlled synthetic mechanism check. Panel (a) verifies that the injected scale imbalance is controlled by the pattern and severity parameters. Panel (b) shows the median conditioning recovery obtained by the four scaling variants and the role-blind flat control on the four imbalance patterns at $S \in \{3, 6\}$. The Base variant is omitted because its reduction is zero by construction; the *none* pattern is omitted from this panel for readability and reported in table 15, where, unlike Base, some rules *worsen* the condition number on that negative control. The coupling-uniform group pools both directions of the shared factor; table 15 splits them. Bars are hatched by variant and outlined with a thin/thick edge (and a lighter/darker fill) for $S = 3/S = 6$, so the grouping reads without color.

The design behaves as intended. For the negative-control pattern *none*, the median pre-scaling condition number is essentially invariant in S , around 1.3×10^3 . When imbalance is injected, κ^{pre} grows with severity only in the affected row classes: at $S = 6$, it reaches about 1.5×10^{14} for *local* and *mixed*, about 4.9×10^8 for *coupling*, and about 3.2×10^7 for *coupling-uniform*. This confirms that the synthetic generator separates local-row, per-row coupling, and collective coupling imbalance by construction (figure 6a).

The shared per-row kernel is validated cleanly. The two variants that normalize every row they reach individually recover the conditioning in every pattern: the full-coverage flat control brings the median κ^{post} to $1.1\text{--}1.3 \times 10^3$ everywhere (about eleven orders of recovery for *local* and *mixed* at $S = 6$), and SA-Mat, whose coupling treatment is also per row, stays within $2.5\text{--}4.7 \times 10^3$ (table 15 and figure 6b). A paired Wilcoxon signed-rank analysis with Benjamini–Hochberg correction gives $p_{\text{BH}} \leq 1.8 \times 10^{-2}$ for every condition-number comparison and $p_{\text{BH}} \leq 4.5 \times 10^{-6}$ for the strong recoveries. One property of this benchmark must be stated explicitly: every injection is a positive row multiplier, so a per-row normalization with full coverage can invert it *by construction*—the flat kernel is the ceiling here, and no role-restricted rule can beat it on the global coefficient range. What the benchmark can separate is how the *structure-aware* designs behave on the rows their taxonomy assigns to the coupling stage.

There the two structure-aware coupling designs separate exactly where the analysis says they should. On *coupling-uniform*, the collective displacement the block-relative stage is designed for, the stage activates and repairs its *oversized* branch: the single factor γ drives the coupling proxy from $\hat{\rho} \approx 10^2\text{--}10^5$ to $\hat{\rho} \approx 1$ and the median κ from 6.9×10^7 to 3.8×10^3 at $S = 6$ on the $\theta > 1$ subgroup. The activation rule is one-sided by design—the stage fires only when the coupling rows are *large* relative to the blocks ($\hat{\rho} > 10$)—so the undersized subgroup is left at base scale ($3.2 \times 10^6 \rightarrow 2.4 \times 10^5$), which the per-row treatments repair (SA-Mat 2.5×10^3 , flat 1.3×10^3). The stage is therefore validated for oversized collective coupling imbalance only; a symmetric or trigger-free variant (letting the minimization of Φ decide the direction) is untested design space, and not trivially safe to adopt: operationally the Simple class sits deep in the undersized regime ($\hat{\rho} \approx 3 \times 10^{-4}$), where the one-sided rule keeps the stage silent and the ablation shows that silence costs nothing there (section 6.8). The *value* of the threshold matters less than its one-sidedness on this benchmark: sweeping the trigger over $\hat{\rho}_{\text{trigger}} \in \{2, 5, 10, 20, 50\}$ leaves the coupling and coupling-uniform cell medians unchanged to the displayed precision. Approximately one instance in twenty falls inside the $[2, 50]$ band, so individual activation decisions may change with the threshold, but no reported cell median changes—the coupling proxy is effectively bimodal (above 10^2 on the oversized instances, below 1 on the undersized ones), so the handful of in-band instances do not move the medians. The one-sided design, not the specific cutoff 10, is what leaves the undersized branch untreated. On the heterogeneous *coupling* pattern the single collective factor cannot compress the spread *among* the coupling rows: with the reliability floor of section 4.4 protecting the individually small rows, SA-Aug recovers partially ($4.9 \times 10^8 \rightarrow 3.3 \times 10^5$ at $S = 6$) while SA-Mat and the flat kernel recover fully (2.5×10^3 and 1.2×10^3): the per-row versus collective design choice, not the metadata itself, is what the coupling patterns discriminate. Two further observations keep the attribution honest. First, the Ruiz variants run inside the same pipeline (their kernel replaces the local stage; the coupling stage is shared), so they track SA-Aug on the coupling patterns and differ only on block rows. Second, the negative control exposes a coverage cost of selectivity: rescaling the local block rows while the coupling rows remain at their natural scale widens the exported range by about an order of magnitude on *none* (SA-Aug $1.3 \times 10^3 \rightarrow 1.3 \times 10^4$), a small but statistically significant worsening that full coverage avoids.

The severe heterogeneous cells also identify a genuine *failure mode* of the common-factor design, which motivated the reliability floor of section 4.4. With the floor disabled, the single-factor variants (SA-Aug and the Ruiz family, which share the coupling stage) fail original-scale verification on 29 of the 1800 runs in the *coupling* and *mixed* cells at $S = 6$, with row violations up to 9.4×10^{-1} and cross-variant objective spreads beyond the MIPGap on 10 of 300 instances; SA-Mat, the flat control, and the base never fail. The mechanism is exactly the ε/d_r mapping of section 6.1: the collective coupling scale is genuinely of order $10^5\text{--}10^6$, so the admitted factor γ is legitimately small, and on the coupling rows that were *individually* small it pushes the scaled row into the regime where the solver’s absolute feasibility tolerance maps back to order one in original units—the solver returns claimed-optimal solutions that violate the original row. A factor fitted to the most extreme rows must not be allowed to determine the scale of rows it does not fit; the per-row treatments never enter this regime here, because they center each row’s coefficient extremes geometrically around one. With the floor enabled (the default), the failure mode is eliminated—all 1800 runs pass verification, the worst row violation is 1.3×10^{-6} , every cross-variant objective spread is below the MIPGap—and the partial recovery on the heterogeneous cells even improves, since the large rows are still repaired while the fragile ones stay at their benign base scale. Because the floor was designed after observing the failure mode on these instances, we validate it out-of-sample: on an independent set of twenty seeds never used in the design, disabling the floor reproduces the failure mode (4 verification failures with violations up to 4.2×10^{-1} and 4 objective disagreements beyond the gap), while *every* floor value in $\{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$ eliminates all failures with comparable conditioning, so the effect is neither an artifact of the calibration seeds nor of the specific threshold 10^{-3} (Table S3, Supporting Information). The synthetic benchmark thus reproduces, under controlled conditions, the reliability mechanism the operational audit measures, and turns it into a designed, independently validated safeguard.

TABLE 16 Solver-internal-scaling baseline (1% MIPGap; the 0.1% sweep is summarized in the text). The unscaled model is run under each solver’s own scaling controls and compared, on the same instances, against the structure-aware variants under default internal scaling. Solver-time sgm ($s = 1$ s) and PAR10, in seconds. The “Base” row is the untransformed model under the solver’s default internal scaling.

Solver	Variant	Simple		Full		SG-Ter-Mer	
		sgm (s)	PAR10 (s)	sgm (s)	PAR10 (s)	sgm (s)	PAR10 (s)
Gurobi	Base (default scaling)	3.94	11.26	109.38	3,448.2	14.09	564.3
	SA-Aug	2.13	2.50	95.24	3,349.7	8.28	554.1
	SA-Mat	2.17	2.56	102.03	3,458.1	8.64	554.6
	Internal, off	41.97	1,541.9	561.95	9,716.1	56.63	740.2
	Internal, equilibration	5.81	20.39	128.45	2,444.7	14.66	566.4
	Internal, geometric	4.40	12.60	138.11	2,497.1	20.91	589.8
	Internal, aggressive	3.79	10.79	133.83	3,469.0	21.48	580.5
CPLEX	Base (default scaling)	2.62	7.24	147.82	5,609.3	2.25	3.57
	SA-Aug	1.73	3.65	127.36	3,616.8	2.40	10.26
	SA-Mat	1.86	4.01	120.88	6,509.4	2.31	5.94
	Internal, none	2.50	3.93	(14.38) ^c	(35.5) ^c	5.17	12.29
	Internal, default (equilibration)	2.55	7.59	159.01	5,621.8	2.23	3.53
	Internal, aggressive	2.43	7.18	189.13	5,645.0	2.18	5.66

Gurobi internal modes are the `ScaleFlag` settings off/equilibration/geometric/aggressive (0/1/2/3); CPLEX modes are `Read::Scale` none/equilibration/aggressive (−1/0/1). Each solver’s default (Gurobi automatic, CPLEX equilibration) is the scaling under which “Base” and the structure-aware variants run; the internal-mode rows override it on the unscaled model. The Full base sgm differs from table 7 (109 vs 83 s): with 34 instances and heavy censoring, the solved-subset geometric mean is sensitive to run-to-run variability (section 8), which is why all comparisons are within-campaign.

^c Fast but objective-inconsistent with the default pipeline: its claimed-optimal objectives disagree with the default pipeline’s beyond the stopping tolerance on 11 of 33 instances (up to 14%, once $6\times$), in both directions; the comparison does not identify which is correct (see the text).

This is the empirical bridge promised in section 3.6, and it answers the natural worry that the idealized propositions—stated for square, nonsingular blocks under exact SVD scaling—say little about a rule built from cheap coefficient proxies on rectangular, inequality-based matrices. The per-row proxy responds to precisely the imbalance the analysis isolates, and the coupling stage realizes the trade-off of proposition 2 on the branch it treats: it drives the block-relative proxy to order one and recovers the conditioning there. The scope of the check is therefore bounded and explicit: the synthetic suite validates the shared per-row normalization kernel under row-multiplicative imbalance, and it exercises the common block-relative coupling factor only on the *oversized* branch of its collective design case; the undersized branch, severe inter-row heterogeneity, and phenomena beyond row-multiplicative imbalance—scale separation between blocks with ill-conditioned interiors, column imbalance, or rows with identical coefficient extremes but different roles—are either untreated by design or not exercised, and are natural extensions of the generator.

Taken together, the four operational scenarios and the controlled synthetic check support a limited but useful interpretation. Under Gurobi, the structured preprocessing rules improve the numerical representation and, under intent-to-treat timing, often reduce solver time or deterministic effort (section 6.1 gives the reliability-adjusted readings). Under CPLEX, the same external transformations do not produce a robust runtime benefit and can be indistinguishable from, or worse than, the base in paired runtime evidence, even when some numerical diagnostics improve. The empirical message is therefore not that one scaling rule always wins. It is that generator-level structure can matter for the numerical representation the solver sees, while whether that translates into performance must be measured through the full solver–model interaction rather than assumed from the metadata (section 6.8).

6.7 | Solver-internal-scaling baseline

The control of section 5.4 that most directly tests the contribution asks whether the solver’s *own* scaling could reproduce the structure-aware effect. Table 16 runs the *unscaled* model under each solver’s internal scaling settings and compares it, on the same instances, against the structure-aware variants run under default internal scaling.

Two facts stand out. First, disabling Gurobi scaling (`ScaleFlag=0`) is catastrophic on Simple and SG-Ter-Mer—Gurobi’s sgm on Simple rises from 3.9 s to 42 s (over the 188 of 204 runs that did not error out, seven of them timing out), and on SG-Ter-Mer to 57 s—which confirms that Gurobi’s default scaling is already a strong, well-chosen baseline rather than a straw man. The CPLEX no-scaling mode (`Read: :Scale=-1`) has mixed behavior: it is near the default on Simple, slower on SG-Ter-Mer, and anomalously fast but objective-inconsistent on Full (below). Second, whether an internal setting matches the external rule must be read per instance, and it is class-dependent. We compare structure-aware against the best Gurobi internal mode on common instances by the median paired ratio $T_{\text{internal}}/T_{\text{SA}}$ (>1 means external is faster) and the matched-pairs rank-biserial r_{rb} . On Simple the two are effectively tied per instance—median ratio 0.97 for SA-Aug ($r_{\text{rb}} = +0.08$) and 0.96 for SA-Mat—so the sgm gap there (SA-Aug 2.13 against the best internal 3.79 s) is a tail effect, not a per-instance advantage. On SG-Ter-Mer the external rule is robustly faster per instance (median ratio 1.17/1.27, $r_{\text{rb}} = +0.59/+0.64$), and on Full it leads more modestly (median ratio 1.17, $r_{\text{rb}} = +0.30/+0.31$). So on the classes where the external rule carries a robust per-instance advantage over the base, the same advantage holds over the solver’s best internal mode; on Simple the internal mode already matches it per instance. The Gurobi gain therefore does not collapse to “any extra scaling pass, and the solver could have produced it” on SG-Ter-Mer and Full, while on Simple retuning the internal scaling does match the external rule instance by instance. This control shows that retuning the solver’s internal scaling on the untransformed model does not substitute for the external preprocessing in the favorable Gurobi classes; it does not test interactions or jointly tuned combinations of external and internal scaling (a small $\{\text{Base, SA – Aug, SA – Mat}\} \times \{\text{internal modes}\}$ factorial would settle that and is left as future work). Which part of the external rule carries the effect—and in particular whether the row-role metadata itself is necessary—is the attribution question examined in [section 6.8](#).

The SG-Ter-Mer class splits by solver. Under CPLEX the comparison is a wash—structure-aware (2.40 s) sits at the base level (2.25 s) and the aggressive internal mode (2.18 s) is marginally ahead—consistent with the rest of the study, where the Gurobi effect, not CPLEX, carries the improvement on this class ([section 6.8](#)). Under Gurobi, by contrast, the win is decisive and no internal mode comes close: structure-aware reaches 8.3 s against a 14.1 s base, the best internal setting (equilibration) only matches the base (14.7 s), and the geometric and aggressive modes are markedly *worse* than the base (20.9 and 21.5 s)—so structure-aware is $1.77\times$ faster than the best internal mode and over $2.5\times$ faster than the worst. The internal-scaling baseline thus supports the central claim where the external rule shows an intent-to-treat Gurobi advantage (Simple and SG-Ter-Mer); under CPLEX it adds a descriptive Simple contrast that the paired tests do not elevate to a robust gain, and, where the rule is not effective (SG-Ter-Mer under CPLEX), it shows structure-aware neither helping nor harming relative to the solver’s tuned internal scaling.

The Full class and the 0.1% sweep complete the picture with two honest qualifications. At 1%, Full behaves like the other classes: the structure-aware sgm (95–102 s) is below the default base (109 s) and below every internal Gurobi mode (128–562 s), although the equilibration and geometric modes save one timeout and therefore lead PAR10. At 0.1%, Simple and SG-Ter-Mer repeat the 1% conclusion (best internal settings 9.4 and 11.8 s against structure-aware 6.2–6.4 and 8.8–9.1 s in this internal-scaling campaign, within run-to-run drift of the 5.4–5.7 s and 8.0–9.0 s reported in [table 8](#)), but Full does not: the retuned internal modes (337–353 s) overtake the external rule (388 s for SA-Aug, base 437 s), so on the hardest class at the strictest tolerance the structure-aware advantage holds over the default pipeline but not over a tuned internal-scaling one. The CPLEX side of the Full sweep contributes a cautionary observation rather than a baseline: disabling internal scaling (`Read: :Scale=-1`) appears dramatically faster (sgm 14.4 against 147.8 s, no timeouts), but its claimed-optimal objectives disagree with the default pipeline’s beyond the stopping tolerance on a third of the instances, in both directions. The available comparison does not identify which configuration is correct—the default pipeline can itself return invalid claimed optima ([section 6.1](#))—so we describe that configuration as objective-inconsistent with the default pipeline, a reminder, in line with the original-scale reliability audit of [section 6.1](#), that on this class scaling choices can change the reported certificate rather than merely the runtime. The remaining CPLEX Full cells at 0.1% are partial (the four-day sweep budget expired) and are omitted.

6.8 | Operational stage attribution

The synthetic benchmark isolates local and coupling imbalance by construction; the operational ablation of [section 5.4](#) does the same on the real hydrothermal classes by running the local-only and coupling-only variants alongside the full architecture under Gurobi. [Table 17](#) reports the result for the three classes at the 1% tolerance (the 0.1% tolerance shows the same ordering in all three classes; in Full, local-only 326 against 326 s for the full rule and coupling-only 376 at the base’s 374 s). In the Full class several instances reach the one-hour limit under every variant, so a raw sgm per variant is computed over that variant’s

TABLE 17 Operational stage ablation under Gurobi (1% MIPGap). The local-row stage reproduces almost the entire effect; the coupling-row stage alone is essentially inert. Solver-time shifted geometric mean (sgm, $s = 1$ s) and PAR10, in seconds.

Class	Variant	sgm (s)	PAR10 (s)
Simple	Base	3.73	10.47
	SA-Aug (local+coupling)	2.02	2.42
	SA-Aug, local only	2.01	2.38
	SA-Aug, coupling only	3.74	10.44
	SA-Mat (local+coupling)	2.04	2.43
	SA-Mat, local only	2.06	2.48
	SA-Mat, coupling only	3.73	10.26
Full	Base	82.55	3,379.7
	SA-Aug (local+coupling)	71.41	3,315.9
	SA-Aug, local only	70.94	3,314.4
	SA-Aug, coupling only	83.69	3,381.5
	SA-Mat (local+coupling)	78.75	4,384.9
	SA-Mat, local only	78.88	4,386.1
	SA-Mat, coupling only	90.21	3,394.1
SG-Ter-Mer	Base	14.42	565.6
	SA-Aug (local+coupling)	8.55	555.1
	SA-Aug, local only	8.58	555.2
	SA-Aug, coupling only	14.45	565.9
	SA-Mat (local+coupling)	8.96	555.6
	SA-Mat, local only	8.83	555.6
	SA-Mat, coupling only	14.43	565.8

“Local only” applies the local-row stage and leaves coupling rows at base scale; “coupling only” applies the coupling-row stage and leaves local rows at base scale. In SG-Ter-Mer, PAR10 in this campaign is dominated by a single near-limit instance that times out under all variants (unlike the main 1% scenario, where SA-Mat solves it within budget), so the attribution here is carried by sgm. In Full, three or four of the 34 instances time out per variant, so PAR10 moves in steps of one timeout and the attribution is likewise carried by sgm over the solved subset.

own solved subset and can mix speed with *which* instances each variant solved. To avoid this, the Full comparisons that carry weight in this paper are paired: the flat-control and stage tables report the median per-instance paired ratio and the matched-pairs r_{tb} over the non-double-timeout pairs (a common set for the two members of each comparison), and the reliability readings of [table 5](#) use PAR10 on the fixed pool with the same timeout penalty for all variants. Where a raw sgm over a solved subset is quoted for Full, it is descriptive and should be read with that caveat.

Before reading these numbers as a stage *ablation*, one fact must be stated: on the operational classes the coupling stage never activates. The augmented coupling proxy is $\hat{\rho} \leq 10$ on every operational instance ($\hat{\rho} \approx 2 \times 10^{-4}$ in Simple, exactly 1 in SG-Ter-Mer and Full; 0 of 197, 0 of 67, 0 of 34 above the threshold), so the stage is skipped, and the exported matrices of the full rule and its local-only reduction are *byte-identical* on the sampled instances of every class. The same holds for SA-Mat: its per-row coupling factors fall in the near-identity band on the well-conditioned balance rows, so its full and local-only exports also coincide byte for byte. This section is therefore an *operational activation audit*, not an ablation: the small sgm differences between full and local-only (3.73 $\rightarrow$ 2.01 against 2.02 in Simple; 82.55 $\rightarrow$ 70.94 against 71.41 in Full; 14.42 $\rightarrow$ 8.58 against 8.55 in SG-Ter-Mer) are run-to-run variability between independent solves of identical models, not evidence about the coupling stage, and the coupling-only variant sits at base level (3.74, 83.69, 14.45) precisely because it changes nothing. The operational Gurobi effect thus comes entirely from local row normalization; the coupling stage’s mechanism is validated only on the synthetic benchmark ([section 6.6](#)), where the proxy does exceed the threshold.

Because the local stage is, in isolation, a per-row geometric-mean normalization of the row’s nonzero coefficient magnitudes—for SA-Aug optionally augmented with the right-hand side, and computable without any row-role information—the natural sharpening of this control is to apply that same kernel to *every* row, with no roles and no coupling stage. [Table 18](#) reports this role-blind variant, run on all three classes. In Simple it is indistinguishable from the local-only rule and from the full architecture (sgm 2.02 at 1%, matching 2.01 and 2.02; the same at 0.1%), so the row-role metadata adds nothing to runtime there. In SG-Ter-Mer it is markedly *faster* than either: sgm 5.25 against 8.38 for both the full rule and local-only at 1% (paired speedup $1.85\times$ versus $1.22\times$, $r_{\text{tb}} = -0.86$), and 5.81 against 9.23 at 0.1%, with the same original-scale verification behavior

as the other variants. In Full the outcome *reverses*: blanket application of the same kernel is slower than the role-informed rule (sgm 85.2 against 71.8 at 1%, paired speedup $1.13\times$ against $1.38\times$; the flat kernel is slower on 21 of 31 optimal pairs, median ratio 1.09, and the direction persists at 0.1%), so on the most integrated class the role-based selection of which rows to scale shows a directionally consistent runtime advantage in the reported campaign. The Full evidence should be read with the appropriate caution: the class has only 34 instances, the heaviest censoring, and 16-thread runs whose run-to-run variability is the largest of the three classes (section 8), so the reversal is a single-campaign, direction-consistent observation rather than a definitive effect size; a seed-replicated confirmation is left as future work.

Exporting the scaled matrices identifies the mechanism precisely, and it is not the coupling treatment. In SG-Ter-Mer the model has 2160 global rows: 360 demand-balance (coupling) rows and 1800 that are global but *not* coupling—chiefly the market shortage-segment rows, whose coefficients span the segment marginal costs (here up to 4000) against a unit shortage-penalty term, with right-hand sides of order 10^5 – 10^7 . The balance rows are already well conditioned ($\hat{\rho} \approx 1.1$), so the coupling stage correctly leaves them unscaled, and the flat kernel leaves them unscaled too (its factor falls in the near-identity band): the coupling rows are *not* where the two rules differ, and the exported balance rows are byte-identical under both. The 1800 shortage rows, however, fall outside both stages of the structure-aware rule—they are neither local-block rows nor coupling rows—so it never scales them, leaving coefficients of order 10^3 and right-hand sides of order 10^6 in the matrix. The flat kernel, scaling every row by $1/\sqrt{M_r m_r}$ over its coefficients *and* right-hand side, geometrically centers their smallest and largest nonzero magnitudes around one, substantially reducing each such row’s scale mismatch; on one SG-Ter-Mer instance that lowers the exported coefficient-and-right-hand-side range ratio—defined as $\max\{|a_{rj}|, |b_r|\} / \min^+\{|a_{rj}|, |b_r|\}$ over nonzero magnitudes, i.e. the ratio of equation (43) extended to include the right-hand side—from 3.7×10^{11} to 1.3×10^{10} , and it is the only difference between the two scaled matrices. The runtime lever under Gurobi is therefore per-row normalization, but *where* to apply it is class-dependent. In SG-Ter-Mer the structure-aware decomposition into local and coupling roles left a coverage gap over exactly the highest-magnitude rows, and the role-blind kernel wins by closing it, without using any metadata. (On the synthetic benchmark the taxonomy leaves no row unclassified, so there is no coverage gap of this kind; the differences there trace to the coupling-stage design instead, section 6.6.) In Full the same control loses, and exporting its matrices sharpens why the reversal is not a matter of *which* rows are scaled: the flat kernel there scales exactly the same two families as in SG-Ter-Mer—the shortage-segment and generation-versus-demand rows—and no rows specific to the nonlinear-hydro Full model (verified by differencing the exported constraints on two instances). The matrix change is thus qualitatively identical across the two classes, yet its runtime effect flips sign. What differs is that in Full those shortage rows are no longer the range-dominating extremes—the nonlinear-hydro height and thermal structure absent from SG-Ter-Mer now sets the coefficient range—so geometrically centering their coefficient and right-hand-side extremes around one is a smaller, and here unfavorable, part of the whole. Why an identical rescaling closes a coverage gap in one model’s branch-and-bound and perturbs another’s is a solver-dynamics question our static matrix analysis cannot resolve; what the control establishes is that neither blanket coverage nor role selectivity dominates across classes, and the trade-off must be measured. The structure-aware rules do improve the solver-facing conditioning diagnostics (table 9), but on these operational classes that gain comes from the *local* stage: in the ablation runs the coupling-only variant leaves the solver-level conditioning ratio at one (recorded in the reproducibility artifacts; it is inert in both conditioning and runtime here), while the coupling stage’s ability to repair a *collective* displacement of the coupling rows is what the controlled synthetic coupling-uniform pattern demonstrates (section 6.6). The coupling metadata drives neither the conditioning improvement nor the runtime gain on the operational testbed; the row-role metadata, used as a *selection filter* for the per-row kernel, has a class-dependent runtime effect—absent in Simple, negative in SG-Ter-Mer, and directionally favorable in the reported Full campaign: an empirical selectivity choice rather than the source of a uniform gain (section 7).

TABLE 18 Role-blind flat control under Gurobi. The same per-row geometric-mean kernel used by the local stage, applied to *every* row with no role or block metadata (single block, no coupling stage), against the full structure-aware rule and the local-only ablation. Solver-time sgm ($s = 1$ s), in seconds; paired speedup versus base and matched-pairs rank-biserial correlation r_{rb} (< 0 faster).

Class	Variant	sgm (s)	Med. paired sp. ^a	r_{rb}
Simple, 1%	SA-Aug (full)	2.02	1.00	−0.23
	SA-Aug, local only	2.01	1.00	−0.23
	Flat (role-blind)	2.02	0.99	−0.23
SG-Ter-Mer, 1%	SA-Aug (full)	8.38	1.22	−0.51
	SA-Aug, local only	8.38	1.22	−0.52
	Flat (role-blind)	5.25	1.85	−0.86
SG-Ter-Mer, 0.1%	SA-Aug (full)	9.23	1.20	−0.54
	SA-Aug, local only	9.22	1.18	−0.53
	Flat (role-blind)	5.81	1.86	−0.84
Full, 1%	SA-Aug (full)	71.77	1.38	−0.34
	SA-Aug, local only	71.84	1.38	−0.34
	Flat (role-blind)	85.23	1.13	−0.16

^a Median of the per-instance paired speedup ratios versus the base; it need not equal the ratio of the displayed shifted geometric means. Effective non-double-timeout pairs: Simple 204 of 204; SG-Ter-Mer 67 of 68 (one double timeout); Full 31–32 of 34 depending on the variant.

The flat control passes the original-scale verification checker at the same rate as the other variants (section 6.1); its speedup is not an artifact of a different or easier optimum. At 0.1% Simple the pattern is the same as at 1% and is omitted; at 0.1% Full the direction also persists over the 30 of 34 instances completed within the campaign budget (sgm 262 for SA-Aug vs 298 for the flat kernel; paired median ratio 1.14, flat slower on 17 of 27 optimal pairs). This is an independent paired batch, so the SA-Aug (full) and local-only sgm differ slightly from table 17 (8.38 here vs 8.55/8.58 s in SG-Ter-Mer) by run-to-run variability (section 8); all comparisons are within-table.

7 | DISCUSSION

The experiments support a bounded claim. Structure-aware positive row scaling changes the solver-facing representation in the intended direction: under Gurobi, intent-to-treat timing shows lower runtime or deterministic effort in the three classes, and after penalizing original-scale verification failures the robust runtime advantage remains clearest in SG-Ter-Mer (section 6.1); under CPLEX 22.1.2.0 the same external scaling yields no robust runtime gain under either reading. The evidence thus separates three issues that are easily conflated: algebraic equivalence, numerical representation, and mixed-integer search performance.

Table 19 keeps the claim bounded: not universal acceleration, but a reproducible way to use block and coupling information before export and a test of whether each solver pipeline benefits from it.

A first objection is that commercial solvers already scale internally—which is exactly why the scaled variants always run against the solver’s default internal scaling and presolve (the internal-scaling control varies those settings only for the unscaled model). The comparison is not external scaling versus a naive solver, but whether a generator can improve the representation before the solver’s own pipeline begins. The mixed Gurobi–CPLEX outcome is informative: the same representation can help one pipeline and be absorbed by another.

A second objection is that any benefit may come from something other than the generator metadata, and it splits into three questions the controls answer separately. First, the choice of *kernel*: the Ruiz baselines swap the geometric-mean kernel for max-norm equilibration inside the same structural pipeline (section 5.3), so they isolate the kernel while holding the selection policy fixed—they do compress coefficient ranges, yet their matched runtime is irregular and does not reliably turn that compression into speed, so the geometric-mean kernel is not interchangeable with a generic one, but this comparison says nothing about the metadata. Second, the choice of *coverage policy*: whether the row-role selection helps is a metadata question, and it is answered only by the role-blind flat control (SA-Aug or local-only versus Flat, section 6.8), whose verdict is class-dependent—neutral in Simple, adverse in SG-Ter-Mer, favorable in Full. Third, whether the solver could have done this itself: the internal-scaling control (section 6.7) shows that on the classes where the external rule is effective, no tuned internal setting reproduces the structure-aware runtime on the same instances (on Full at the strictest tolerance the tuned internal modes do overtake it). The metadata’s value is therefore attributed strictly to the second comparison, not to the Ruiz contrast, which concerns the kernel alone.

TABLE 19 Claim, evidence, and limitation are distinct parts of the argument.

Claim	Evidence in this study	Limitation
Algebraic equivalence	Positive factors multiply complete constraints; the claim is proved analytically. The post-solve original-scale checks are separate evidence of <i>numerical reliability</i> of the returned solutions, not of equivalence.	This claim concerns the mathematical MIP, not the path followed by branch-and-bound nor finite-precision behavior.
Numerical representation	Structure-aware variants are associated with lower path-specific fixed-incumbent LP conditioning diagnostics in the harder classes.	These diagnostics fix each variant's own incumbent (not one common LP) and describe representation; they do not prove faster MIP search.
Gurobi performance	Under intent-to-treat timing, per-row scaling reduces runtime or deterministic-effort metrics across the three classes in both Gurobi scenarios.	The lever is the per-row kernel; the role metadata's runtime effect is class-dependent (a role-blind flat control ties in Simple, wins in SG-Ter-Mer, and is directionally slower in the reported Full campaign, table 18); and after penalizing original-scale verification failures the robust runtime advantage remains clearest in SG-Ter-Mer and does not survive in Simple or Full (section 6.1).
Solver dependence	CPLEX remains near parity or deteriorates in paired runtime evidence, even when some numerical diagnostics improve.	The result is tied to CPLEX 22.1.2.0, default settings, and the ClusterUY environment.
Generalization	Generator-level block information can guide numerical preprocessing before export; the synthetic benchmark injects local and coupling imbalance by construction, validates the per-row kernel throughout, and exercises the coupling stage on the oversized branch of its collective design case.	Runtime evidence still comes from one hydrothermal dispatch platform; the synthetic benchmark is a mechanism check restricted to row-multiplicative imbalance, not a performance benchmark.

The table keeps the main characters of the argument visible: the algebraic transformation, the numerical representation, the solver pipeline, and the scope of the evidence.

A mechanistic caveat must accompany the conditioning evidence. If better conditioning drove faster search, the best-conditioned variant would be fastest, but the data do not line up that way: the augmented rule improves solver-level conditioning more than the matrix-only rule, yet the matrix-only rule wins runtime more often in Simple and SG-Ter-Mer, and the association between the conditioning ratio and speedup is weak (a small Spearman correlation explaining a few percent of variance). The honest reading is that conditioning is at best a partial mediator: the representation change is real and in the intended direction, but the runtime benefit is plausibly routed through mechanisms the metric does not capture—which presolve reductions fire, or how many simplex bound-flips and node relaxations the scaled model induces. This reading has precedent: large-scale LP experiments find the runtime effect of scaling irregular even where conditioning improves¹⁵, and the conditioning of the LP relaxations is known to evolve along the branch-and-cut tree in ways a static snapshot of the exported matrix cannot summarize²⁰. We therefore present conditioning as evidence that the representation improved, not as the proven cause of the speedup. The role-blind flat control ([table 18](#)) makes the dissociation sharpest: applying the plain per-row kernel to every row, with no metadata, matches the structure-aware rule in Simple and beats it in SG-Ter-Mer—while in the reported Full campaign the role-informed selection shows a directionally consistent runtime edge over the blanket kernel—so the rule that most improves conditioning is not the one that solves fastest, and the runtime accelerator is the cheap per-row normalization itself, with the row roles acting only as a class-dependent selection filter. The structure-aware rules do lower the solver-level conditioning ratios below the Ruiz and unscaled baselines; but that gain flows through the local stage, whose kernel is itself role-blind, and the synthetic benchmark bounds the claim exactly: there the full-coverage flat kernel matches or beats every structure-aware variant on the global coefficient range—as it must, since every injection is a positive row multiplier—while the coupling stage's distinctive repair is confined to the oversized branch of its collective design case ([section 6.6](#)). The coupling metadata make the collective-factor intervention *possible*; the benchmark validates that intervention only for oversized uniform coupling displacement, and it does not establish a general conditioning advantage of metadata over full-coverage per-row scaling. The metadata's operational contribution is *selection*—where to apply the kernel—and becomes runtime-visible only where the taxonomy's coverage genuinely differs from full coverage ([table 18](#)).

A third objection is that the right-hand side may be unnecessary, and the clean evidence comes from the local-only pair of the stage ablation, whose members differ *only* in whether b enters the per-row extremes: the coefficient-only version remains close in Simple and SG-Ter-Mer (2.06 vs 2.01 s and 8.83 vs 8.58 s, within 2–3%) and recovers most of the improvement in Full, although the augmented rule retains an approximately 11% solver-time advantage there (78.9 vs 70.9 s). Most of the benefit therefore does not require the right-hand side, with the largest residual advantage of including it in Full. The head-to-head SA-Aug–SA-Mat comparison is consistent with this but confounds the right-hand side with the coupling criterion ([section 4.3](#)),

so we do not read it as an ablation. Other generated models may differ, but the present gains largely do not require b in the diagnosis.

A fourth objection comes from CPLEX: several transformations improve static or solver-level diagnostics, yet paired runtime evidence stays neutral to unfavorable. This does not weaken the numerical claim; it limits the performance claim. A commercial MIP run is not a linear-algebra experiment—presolve, internal scaling, relaxation bases, cuts, incumbents, branching, heuristics, and tolerances all stand between the exported matrix and the runtime—so external scaling may improve one stage while the binding bottleneck lies elsewhere. We probed the most natural such candidate—that CPLEX’s presolve neutralizes the external scaling—by re-running the Simple class under CPLEX with presolve disabled. The per-instance effect stayed essentially neutral in both regimes (median paired speedup ≈ 1.00 , with the structure-aware variant faster on about half of the 204 instances), so the absence of a robust CPLEX gain is not a presolve artifact: it persists with presolve on or off. We therefore report the CPLEX neutrality as an intrinsic, scoped finding rather than an effect that a different internal configuration would recover.

The methodological lesson: numerical preprocessing for MIPs should be judged on four separable layers—algebraic equivalence, numerical diagnostics, matched runtime, and solver contrast—and never on one coefficient-range statistic or one runtime table. The study reports all four layers and keeps their claims separate.

The controlled synthetic benchmark closes part of the gap: when the source of imbalance is known by construction, the variants respond as intended at the level of conditioning, and with the reliability floor in place original-scale feasibility and objective agreement hold across the grid; the same severe heterogeneous coupling cells expose, with the floor disabled, the numerical-reliability failure mode of the common-factor variants (section 6.6). It does not close the performance gap—the synthetic instances solve in milliseconds—so runtime claims still come from operational instances, and the operational stage ablation (section 5.4) separates local-row normalization from the coupling-row treatment on real data. Other domains (logistics, production planning, network design) would further test whether the mechanism persists as the generator structure changes; it should transfer only insofar as it shares the premise of sparse global coupling over local blocks.

7.1 | Deployability and downstream effects

The method presupposes that the metadata it needs—row role, block ownership, and coupling incidence—are available at model-generation time. This is immediate in a generator that the team owns, as here, but it is not free in the mainstream algebraic modeling languages (Pyomo, JuMP, GAMS, AMPL) that many practitioners use, which typically export a flat matrix and do not surface component structure. Adoption therefore divides into two cases. A team that owns its generator can emit the three metadata fields at low cost, since the generator already iterates over components when it builds the rows. A team that does not own its generator would need either to instrument the export path or to *recover* block structure after export, for example by hypergraph or matrix partitioning of the kind used by automatic Dantzig–Wolfe structure detectors²¹; such a fallback widens applicability beyond owned generators at the price of an approximate, detected block map. The practical value proposition is correspondingly narrow and concrete: for a Gurobi user running many hard instances of a generated model whose structure is known, the method can reduce aggregate runtime and may occasionally rescue a near-limit instance at near-zero marginal cost—though the timeout evidence in this study is not stable across campaigns (the SG-Ter-Mer PAR10 advantage rests on a single timeout-rescued instance that the stage-ablation and internal-scaling campaigns do not reproduce), so the robust signal is the shifted geometric mean, not the timeout count—with no benefit demonstrated for CPLEX and no per-instance speedup promised on easy instances.

A second deployment caveat concerns quantities other than the optimal point. If the original row $a_r^\top x \leq b_r$ is multiplied by $d_r > 0$, the dual multiplier the solver reports for the scaled row satisfies $\pi_r^{\text{scaled}} = \pi_r^{\text{original}}/d_r$, so the original-scale value is recovered as

$$\pi_r^{\text{original}} = d_r \pi_r^{\text{scaled}}. \quad (49)$$

Operations workflows that read shadow prices—marginal costs of demand balances, water values, reserve prices—must apply equation (49) with the recorded row factors before interpreting them (the primal variables, by contrast, need no transformation under row-only scaling). Because the row factors are stored for the audit trail, the recovery is mechanical; but it must be applied, or downstream dual interpretation will be off by the per-row scale.

8 | STRENGTHS, LIMITATIONS, AND REPRODUCIBILITY

The study is strongest where the evidence is matched: runtime comparisons use the same instance, solver, and tolerance for base and variants; numerical diagnostics are reported separately from speed; and feasibility and integrality are checked in the original scale. These choices keep the argument auditable—a reader can see which evidence supports algebraic equivalence, which supports numerical improvement, and which supports solver-dependent acceleration.

The main limitation is domain coverage. The method applies to generated modular MIPs with identifiable local blocks and coupling rows, but the evidence comes from one hydrothermal dispatch platform. That platform supplies a realistic and numerically heterogeneous testbed. It does not prove that the same behavior will appear in other generated optimization domains.

A second limitation is component-level attribution within the operational testbed. The core comparison evaluates the full local–block–coupling architecture against right-hand-side and generic-scaling alternatives; the operational stage ablation introduced in [section 5.4](#) adds the local-only and coupling-only variants on the real hydrothermal classes specifically to localize the effect to a stage. To the extent that the ablation cannot cleanly separate the stages—because the stages interact, the coupling-row factor being defined relative to the already-scaled blocks—the residual limitation is one of attribution, not of validity: every variant is still an exact reformulation of the same MIP, so the solver-dependent performance pattern stands regardless of how the credit splits between stages.

A third limitation is the choice of the row as the unit of intervention. The method scales complete rows, which keeps the transformation trivially equivalence-preserving and leaves variable bounds and integrality untouched. It does not, however, test the competing hypothesis that *column*-scale heterogeneity—water volumes, flows, costs, and power mixed across variables—is the larger numerical lever in these models. Generic column scaling is included only inside the Ruiz+Cols baseline, and the synthetic generator injects row imbalance by construction, so it cannot, by design, reveal whether a structured *column*-scaling counterpart would help more. The row-only framing is thus a deliberate scoping decision, not an established claim that row imbalance dominates; a structure-aware column-scaling rule is a natural and untested companion to the present work.

Solver dependence is another threat. With internal presolve and scaling left enabled, the measured effect is that of an external layer inside realistic commercial pipelines—relevant to practice, but conditional on solver version, settings, and environment. The negative CPLEX evidence is therefore a finding about the tested pipeline, not a universal statement about CPLEX.

A related threat is run-to-run performance variability. Each instance–variant–scenario cell is measured by a single run under default solver seeds, so the well-documented variability of MIP performance is not averaged out. The repeated base campaigns bound it empirically: across the main scenario, the internal-scaling baseline, and the stage ablation ([tables 7, 16 and 17](#)), the Simple base sgm spans 3.73–4.05 s and the SG-Ter-Mer base 14.09–14.42 s—a run-to-run drift of under 10%, an order of magnitude smaller than the $1.8\text{--}2\times$ structure-aware effects, and the deterministic-effort metrics (Work units, ticks) are insensitive to it. The Full class drifts more (base sgm 82.5–109.4 s across the same campaigns): with 34 instances and heavy censoring, its solved-subset geometric mean is fragile, which is why every Full conclusion in this paper rests on within-campaign matched pairs rather than on cross-campaign level comparisons. Seed-replicated runs would nonetheless be a stronger control and are left as future work.

The numerical diagnostics have limited reach: exported coefficient ranges are static matrix summaries, and solver-level conditioning depends on the state the solver exposes and on auxiliary fixed-integer LP computations; neither measures full MIP difficulty. The paper therefore reports original-scale feasibility, integrality in original variable units, objective checks, performance profiles, deterministic effort, paired tests, and primal–dual progress as complementary evidence.

A related limitation is that the selective rule carries no global acceptance test: it can degrade an already well-scaled representation even when every individual factor passes the per-factor safeguards, as the synthetic negative control shows (SA-Aug widens the exported range about tenfold on *none*, [table 15](#)). An acceptance rule—scale provisionally, recompute the diagnostic, keep the transformation only if it does not worsen—would prevent this by construction, but it is not innocuous to bolt on: keyed to the exported range, it would have rejected the scaling on roughly a third of the operational structured runs, many of which nevertheless ran faster, precisely because the exported range is a weak predictor of runtime ([table 11](#)). Designing an acceptance diagnostic that tracks the runtime-relevant effect is left open.

Finally, the experiments use operational data whose raw inputs may not be fully redistributable, so the reproducibility package includes public code, aggregate results, paired-instance identifiers, processed logs, and rebuild scripts. The controlled synthetic benchmark ([sections 5.2 and 6.6](#)) is an auditable complement when raw data cannot be released: it reproduces the numerical mechanism without claiming operational speedup.

9 | CONCLUSION

Generated MIPs contain useful structure before export—local rows, component blocks, and coupling rows—that the solver usually never sees beyond the flat matrix. This paper uses that pre-export information to choose positive row scalings that preserve the optimization problem but change its numerical representation.

The answer to the research question is qualified but affirmative: structure-aware positive row scaling lowers path-specific solver-facing numerical diagnostics for the tested block-structured hydrothermal MIPs, and under Gurobi, intent-to-treat timing shows lower runtime or deterministic-effort metrics in the three classes—after penalizing runs that fail original-scale verification, the robust runtime advantage remains clearest in SG-Ter-Mer and does not survive in Simple or Full—whereas under CPLEX 22.1.2.0 there is no robust gain under either reading. The method changed the representation in the intended direction on the operational testbed; acceleration remains solver-dependent.

The contribution divides along the same line the evidence does. The local–block–coupling architecture—scale local rows first, summarize blocks second, scale coupling rows last with the scales of the blocks they connect—uses information the generator has but a flat-matrix routine may not, and, because each factor multiplies a complete constraint by a strictly positive scalar, feasible points, objective values, and integer restrictions are unchanged. On the *numerical* side the augmented rule lowers the solver-facing conditioning diagnostics below the unscaled model and the Ruiz-kernel baselines on the operational testbed—though it has no general conditioning advantage over the role-blind flat control (the matrix-only rule does so in Simple and Full and stays comparable to Ruiz in SG-Ter-Mer); that gain flows through the role-blind per-row kernel—on the synthetic benchmark the full-coverage flat control matches or beats every structure-aware variant on the global range, and the coupling stage’s distinctive repair is confined to its collective design case—so the metadata claims no general conditioning advantage of its own. On the *runtime* side the attribution is more sober. The stage ablation locates the Gurobi speedup in the local stage, and the role-blind flat control (table 18) shows that the row roles act as a class-dependent selection filter for the same per-row kernel: applied to every row with no metadata, the kernel matches the structure-aware rule in Simple, is substantially faster in SG-Ter-Mer—the role-based rule left a coverage gap over the high-magnitude global rows that belong to no local block and are not coupling rows, exactly the rows the flat kernel reaches—and is slower in Full, where the role-informed selection shows a directionally consistent edge in the reported single campaign (which, given the sample size, censoring, and thread-level variability, we do not read as a definitive effect size). The practical accelerator is therefore the cheap per-row normalization itself; the metadata does not produce a uniform runtime gain, but deciding *where* to apply the kernel is an empirical choice per class that the flat control makes measurable. Establishing even this bounded result required separating four kinds of evidence—algebraic equivalence, original-scale feasibility, solver-facing diagnostics, and matched per-solver runtime—which is as much a methodological takeaway as the scaling rule itself: the rule that best conditions the matrix is not the rule that solves fastest, and only matched per-solver runtime tells them apart.

Future work should test the mechanism beyond hydrothermal dispatch—larger synthetic suites and applications in logistics, production planning, and network design would show whether the pattern persists as the engineering structure changes—and should trace, inside the solver, why the *same* rescaling that closes the SG-Ter-Mer coverage gap slows the Full branch-and-bound (section 6.8 shows the flat kernel scales the same rows in both classes, so the reversal is a solver-dynamics effect our static analysis cannot resolve). Two concrete lessons for structure-driven preprocessing already follow: a rule that scales only the rows it recognizes by role can silently miss the worst-scaled rows, and a rule that scales every row indiscriminately can overshoot on integrated models—so neither completeness of coverage nor role sophistication should be assumed; a practical scaling layer should measure both, exactly as the role-blind control does here.

AUTHOR CONTRIBUTIONS

Mathias Rodríguez Castro: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review and editing.

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The author declares no conflicts of interest.

ETHICS STATEMENT

This study consists of computational experiments and does not involve human participants, animal subjects, patient data, or clinical-trial data.

DATA AVAILABILITY STATEMENT

The reproducibility package for this study is publicly available at <https://github.com/MathiasRodriguezCastro/structure-aware-row-scaling> and archived on Zenodo at <https://doi.org/10.5281/zenodo.20648950>. The repository contains the cleaned experimental source code, preprocessing scripts, all redistributable dispatch input instances and the synthetic benchmark instances, the aggregate result tables and processed representations required to reproduce the reported aggregates, paired-instance identifiers, and the scripts used to rebuild the reported tables and figures. Some original raw operational inputs cannot be redistributed; the affected runs are documented through paired identifiers, processed logs, and aggregate outputs, so their figures reproduce from the package even where the raw input is withheld. Large per-instance solver logs and bulky dispatch CSVs are distributed as external archives linked from the repository's `artifacts/README.md`. The repository also documents the solver versions, scenario definitions, command-line options, time limits, MIPGap settings, preprocessing flags, and log-parsing rules used to construct the matched comparisons. The aggregate results were produced by the original ClusterUY runs, while the released C++ code is a cleaned reproducibility version containing the monolithic MIP path and preprocessing methods used in the paper.

SUPPORTING INFORMATION

A Supporting Information document accompanies this article with the 0.1%-MIPGap Gurobi and CPLEX distributional, diagnostic, and performance-profile views (Figures S1–S4) and four tables: the hyperparameter sensitivity of the structure-aware rule (Table S1), the per-variant original-scale reliability breakdown with PAR10 under the three readings of section 6.1 (Table S2), the out-of-sample sensitivity of the collective coupling reliability floor over three orders of magnitude on independent seeds (Table S3), and the evaluation of the optional local reliability floor on the failing Simple subset (Table S4). Additional processed logs, extended tables, and diagnostic plots are available in the reproducibility repository.

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